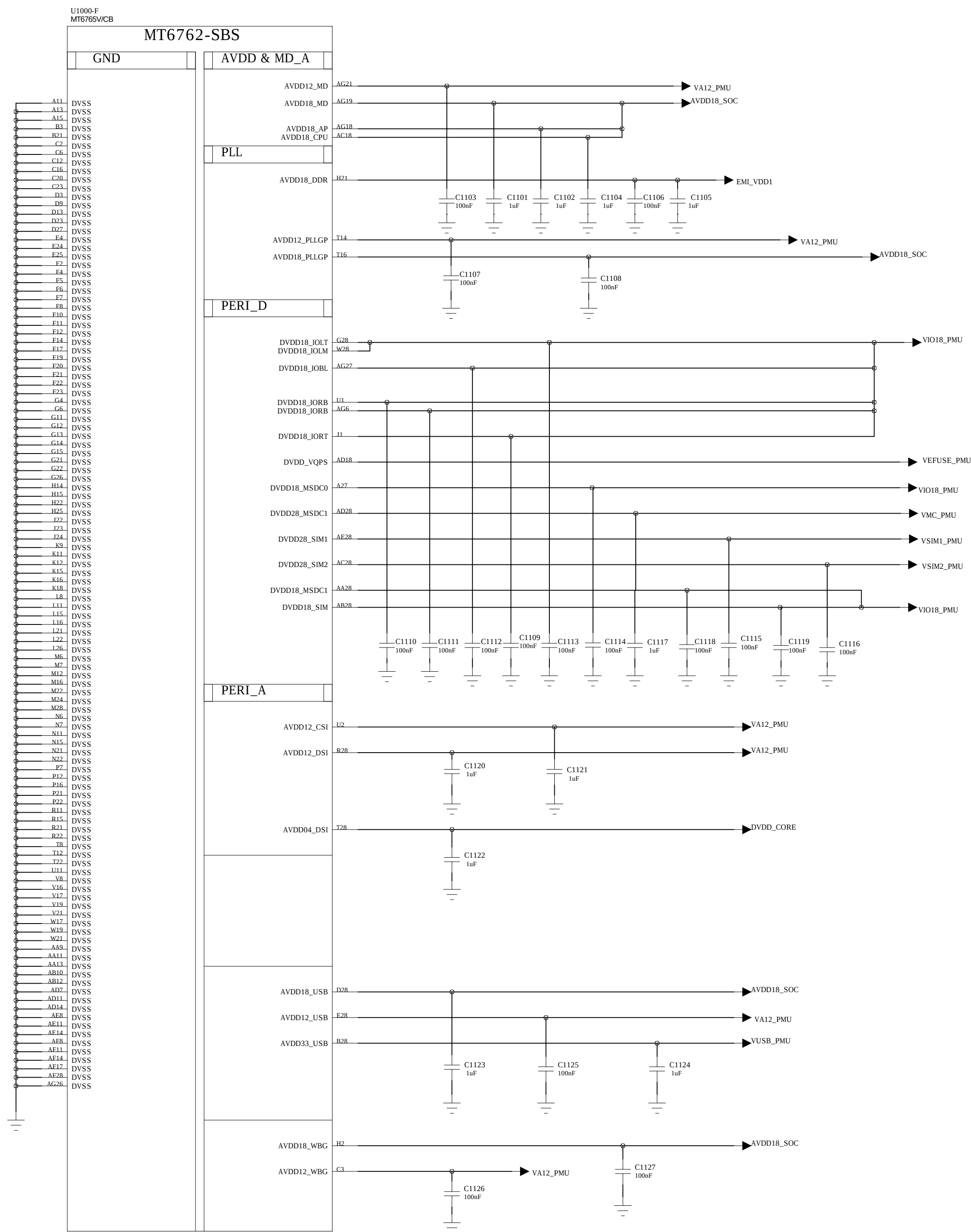


REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Schematic design notice of "11_BB_POWER_IO" page.

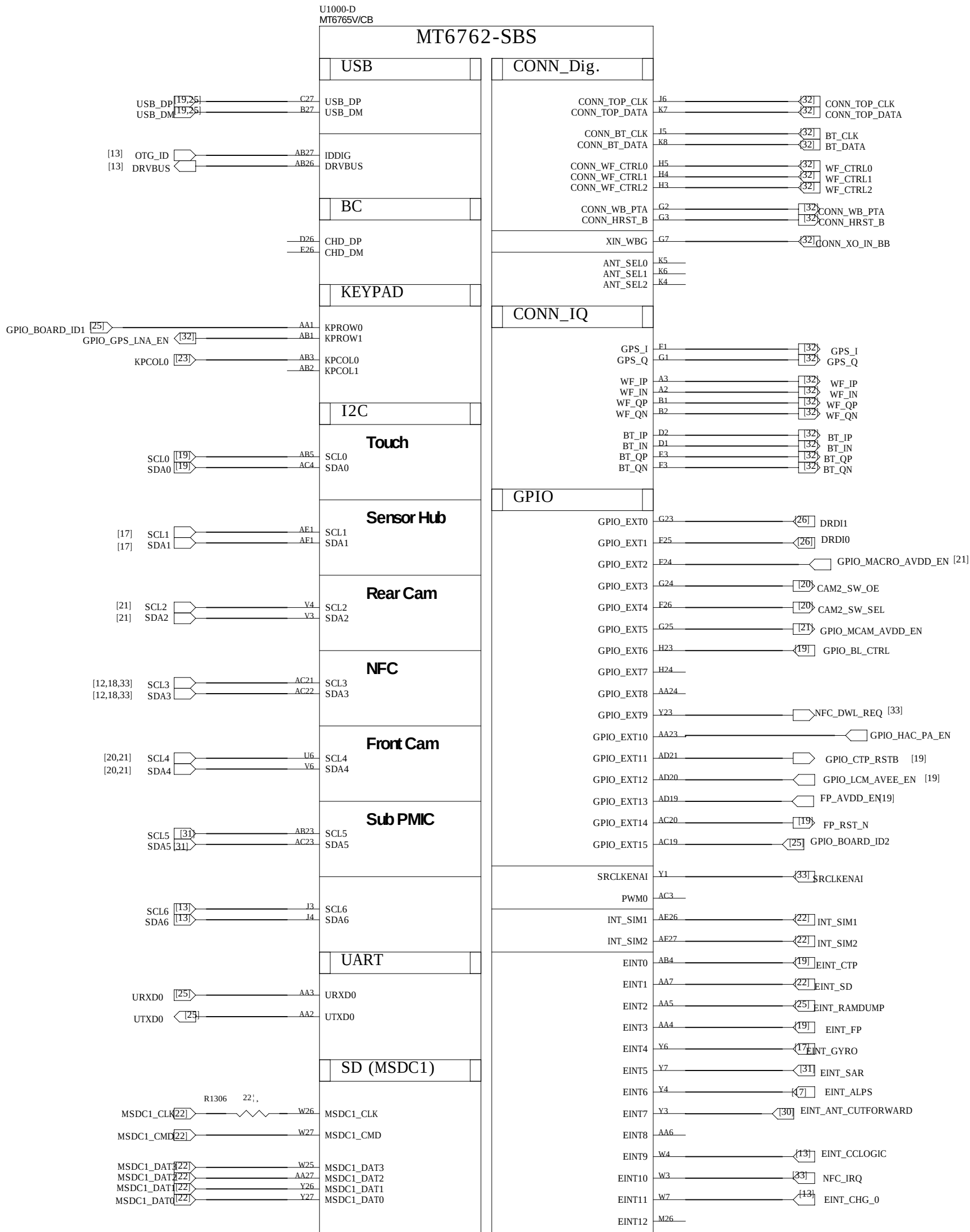
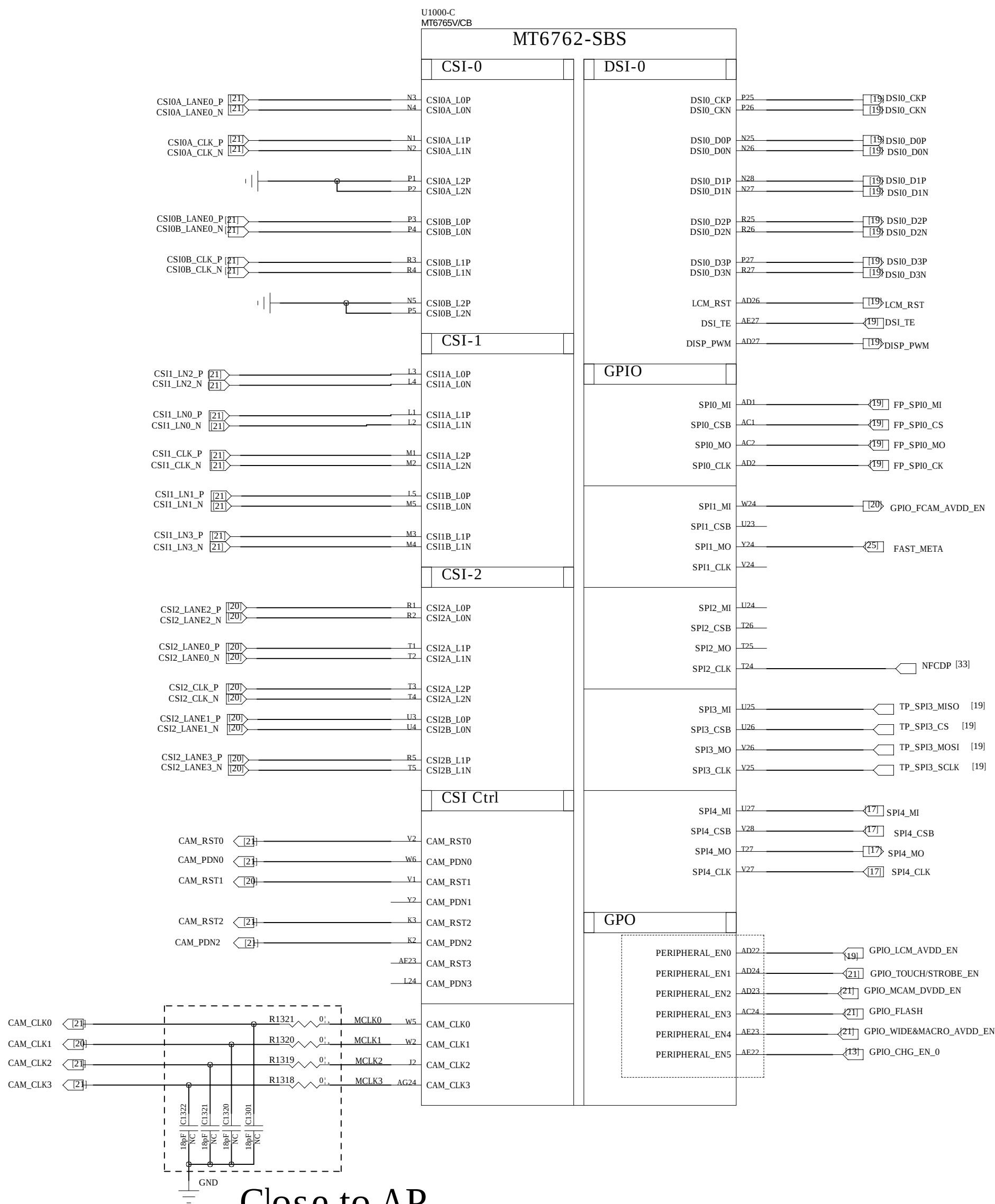
Note 11-1: C4101 closed DVDD18_MSDC0 150mil

Note 11-2: C4302 closed DVDD28_MSDC1 150mil

Note 11-3: C4301 closed DVDD18_MSDC1 150mil

		COMPANY: Tino			
		TITLE: MT6765			
		CODE: P410	SIZE: D	DRAWING NO: P410	REV: V1.0
DRAWN: Marshall	DATED: 2018-09-22A				
CHECKED: <Checked By>	DATED: <Checked Date>				
QUALITY CONTROL: <QC By>	DATED: <QC Date>				
RELEASED: <Released By>	DATED: <Release Date>				
		SCALE: <Scale>		SHEET: 06 33	

REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE



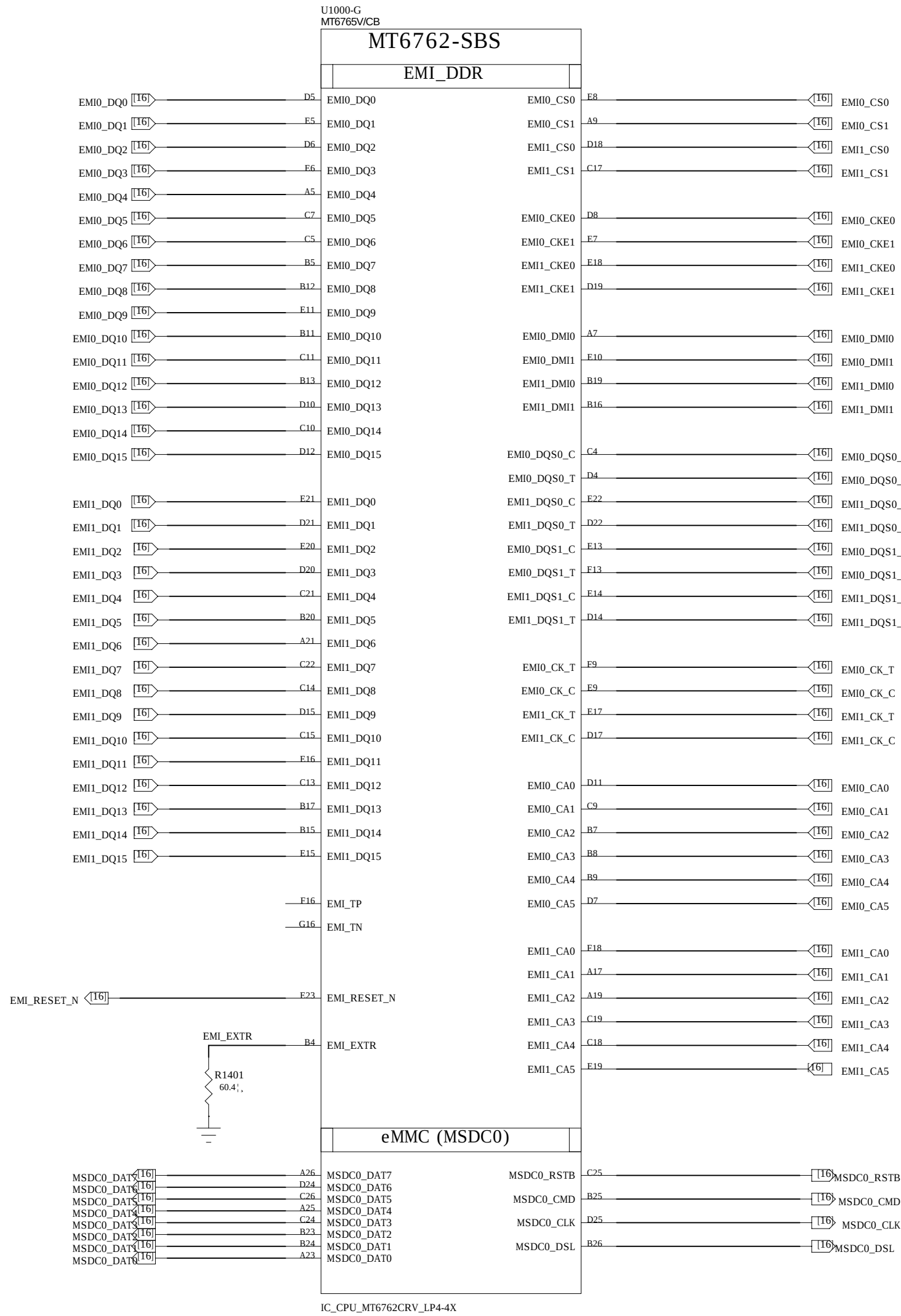
Schematic design notice of "13_BB_2" page.

Note 13-1: The enable pin of acoustic or optoelectronic devices (e.g. SPK AMP/Backlight/Charger OCP/OVP) suggest to use Peripheral_EN[0:5]
If use other GPIOs as enable pin, suggest to reserve 0201 NC to GND

DRAWN: Marshall	DATED: 2018-09-22A
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

COMPANY: Tinno			
TITLE: MT6765			
CODE: P410	SIZE: D	DRAWING NO: P410	REV: V1.0
SCALE: <Scale>		SHEET 5f 33	

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



Schematic design notice of "14_BB_3" page.

Note 14-1: R4001 please select 60.4 ohm (1%) resistor

Note 14-2: Please check eMCP LP3 and eMCP LP4X/LP4 pin mux

COMPANY: Tinno			
TITLE: MT6765			
DRAWN: Marshall	DATED: 2018-09-22A	CODE: P410	SIZE: D
CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO: P410	REV: V1.0
QUALITY CONTROL: <QC By>	DATED: <QC Date>	SCALE: <Scale>	
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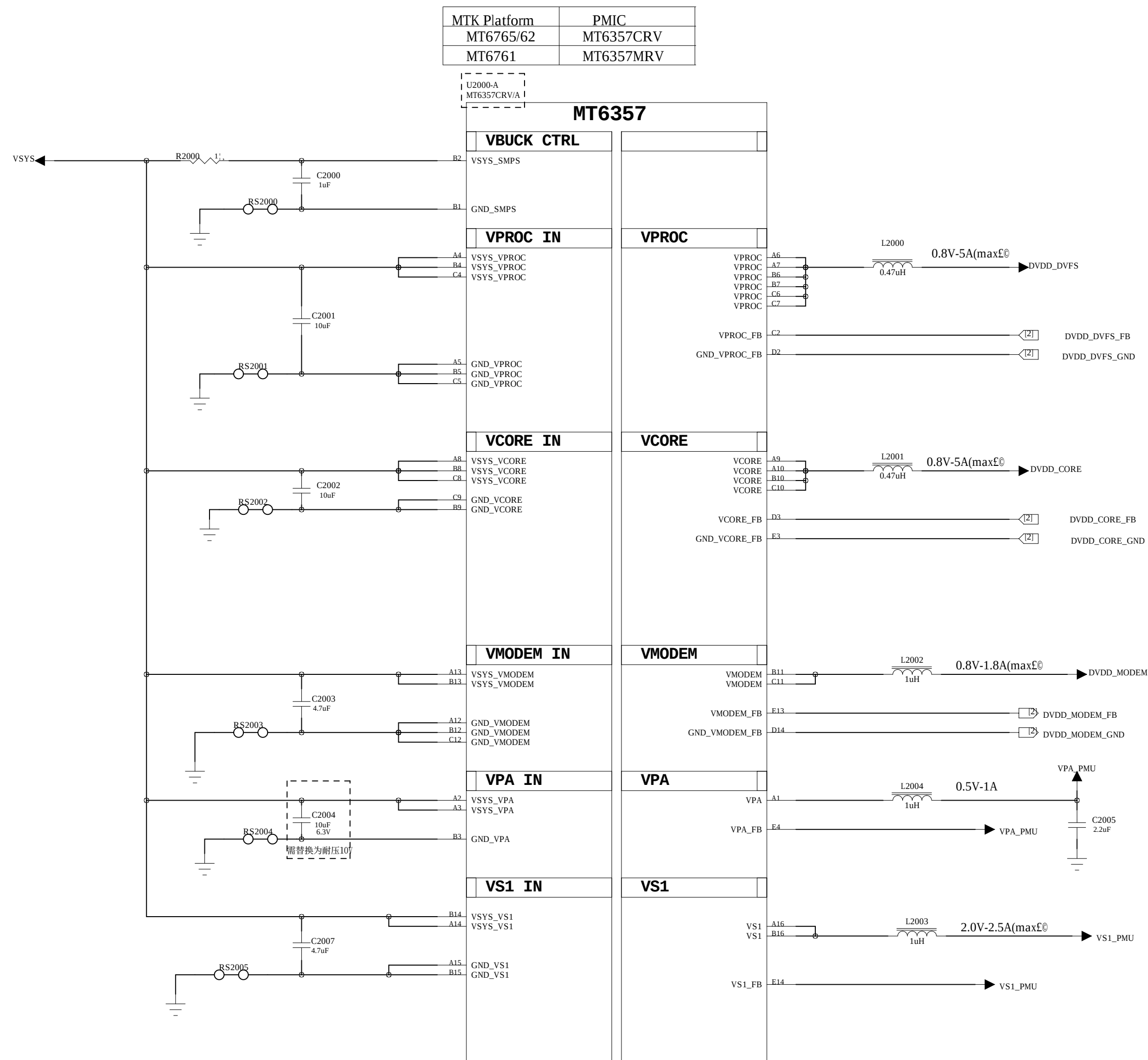
Schematic design notice of "20_POWER_MT6357_Buck"

Note 20-1: C2005, please choose 0402 size

Note 20-2: PMIC Part number notice for MT6765/62/61 platform

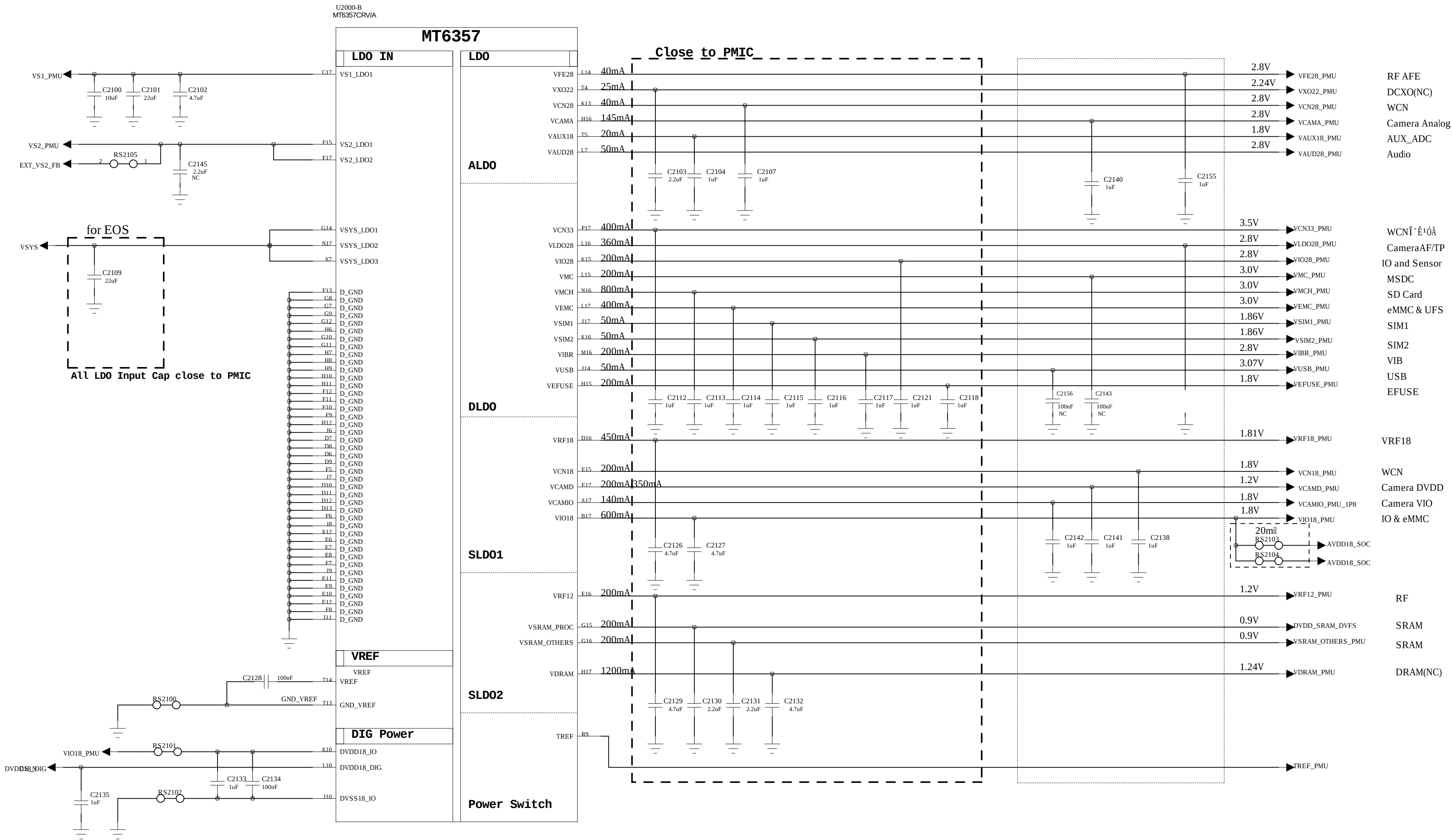
MTK Platform	PMIC
MT6765 / 62	MT6357 CRV
MT6761	MT6357 MRV

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



DRAWN: Marshall		DATED: 2018-09-22A		M16/65			
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SCALE: <Scale>				SHEET: 86		33	

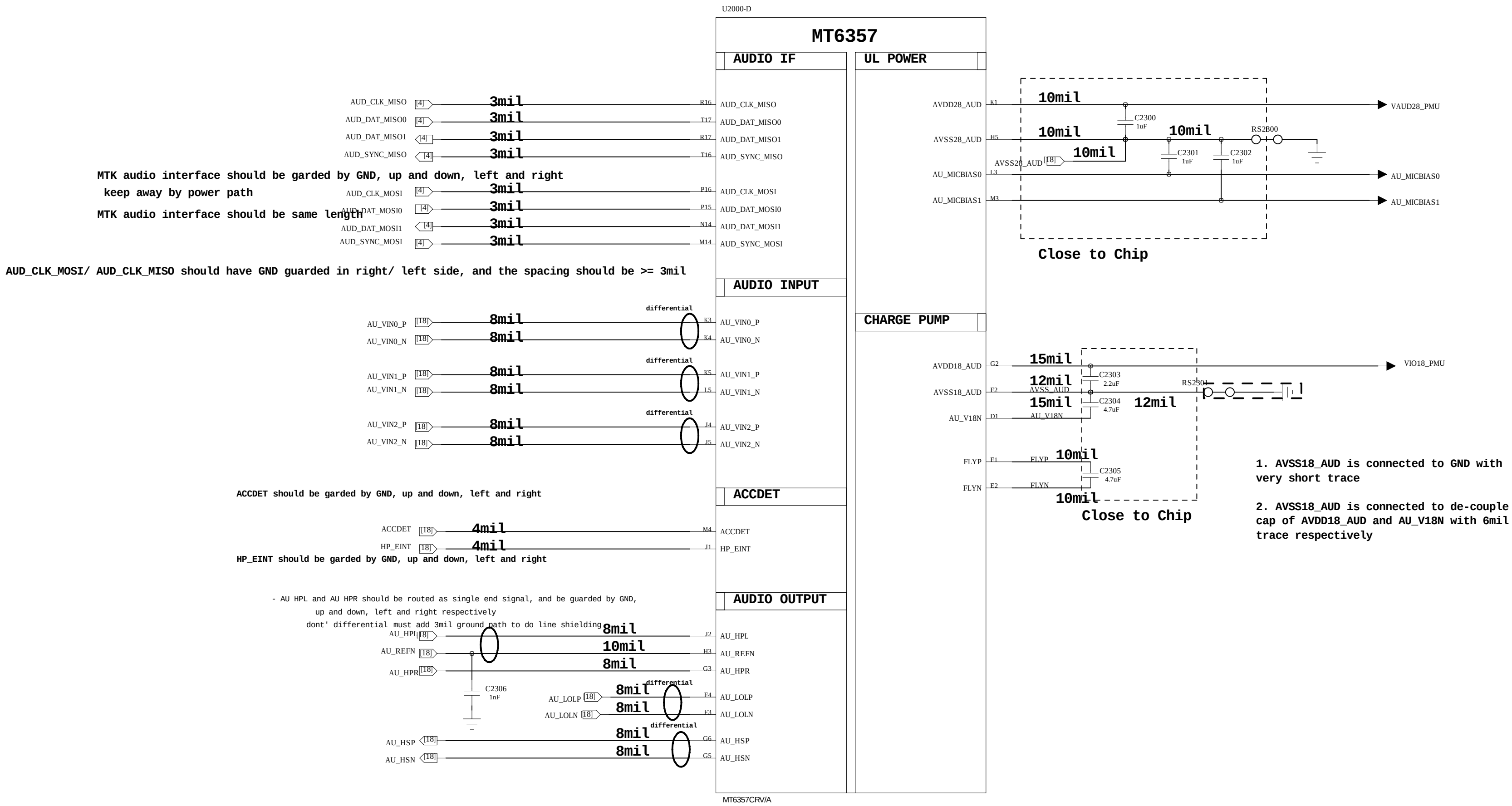
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: Tinno			
TITLE: MT6765			
CODE: P410	SIZE: D	DRAWING NO: P410	REV: V1.0
SCALE: <Scale>		SHEET: 9 33	

DRAWN: Marshall	DATED: 2018-09-22A
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

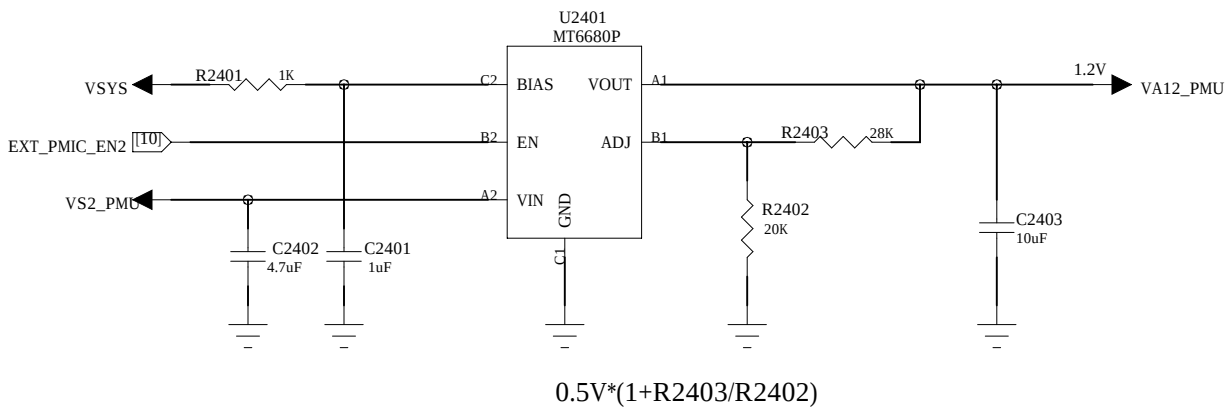
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: Tinno			
TITLE: MT6765			
DRAWN: Marshall	DATED: 2018-09-22A	CODE: P410	SIZE: D
CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO: P410	REV: V1.0
QUALITY CONTROL: <QC By>	DATED: <QC Date>	SCALE: <Scale>	
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REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

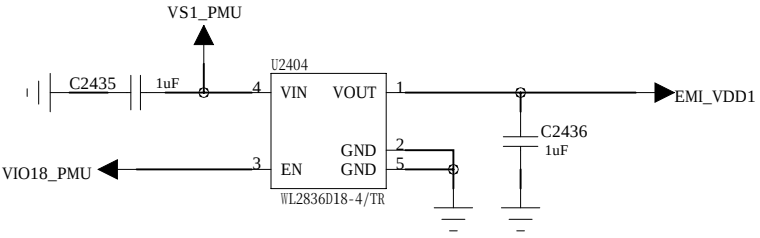
LDO for VA12



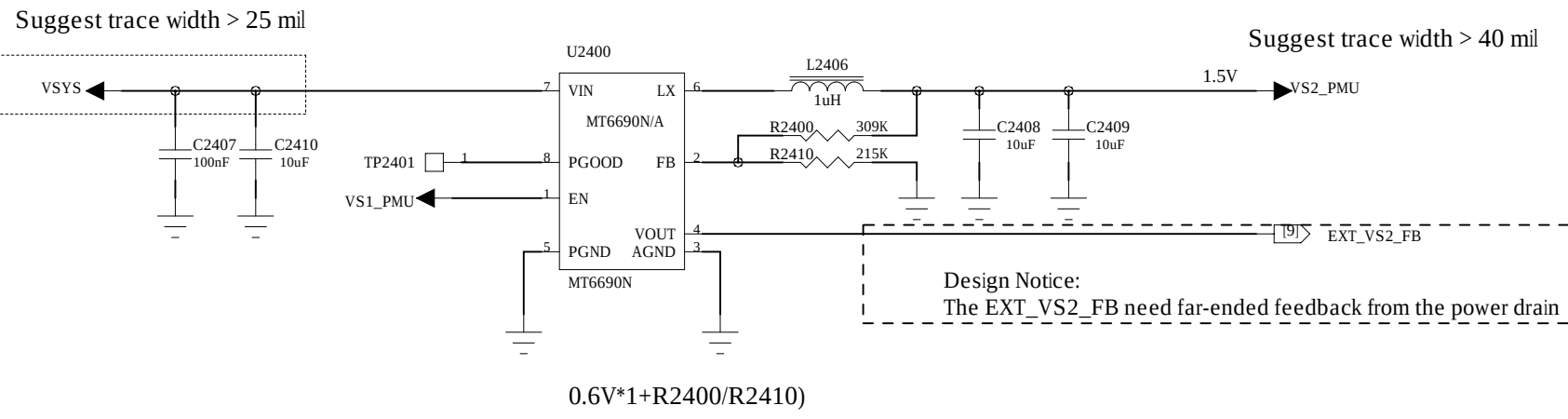
$0.5V \times (1 + R2403/R2402)$

LPDDR4X VDD1 1.8V LDO

Suggest trace width > 20 mil



Ext. Bulk for VS2
For Low Power Performance

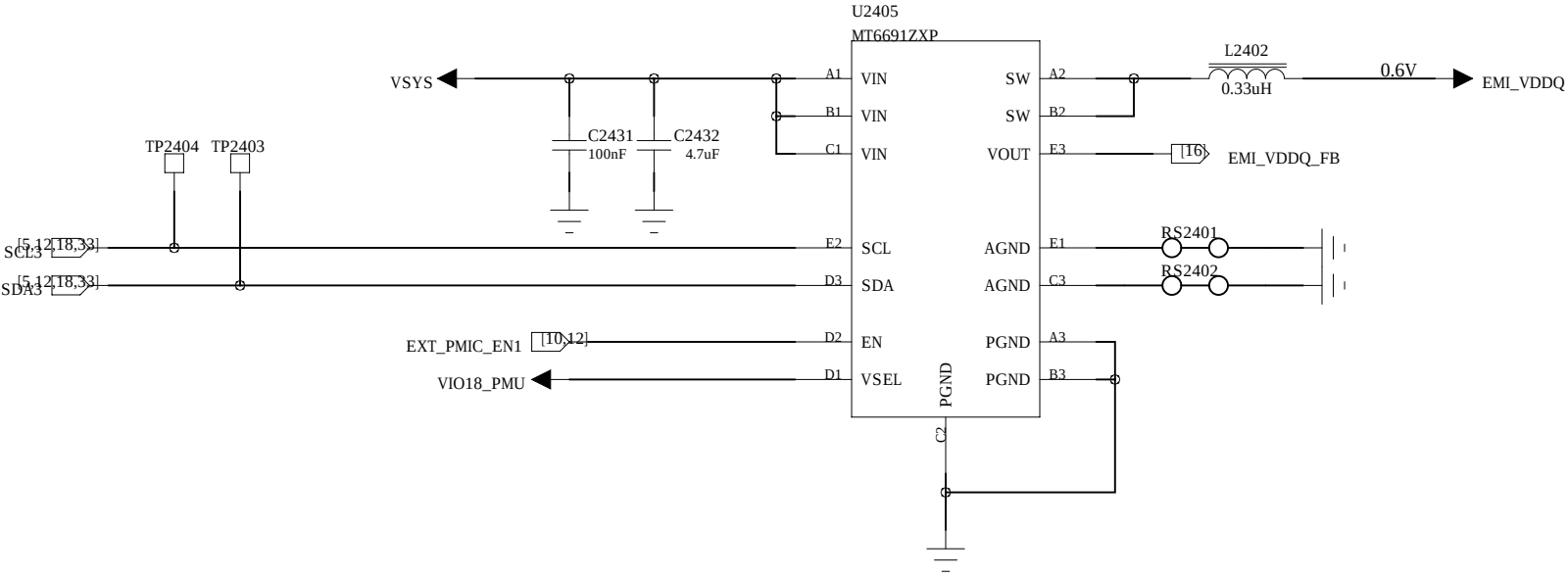


$0.6V \times (1 + R2400/R2410)$

Ext. buck LP4/4X VDDQ 1.125V/0.6V

If use LPDDR4, NC U2802 and peripheral components
MT6691ZXP/A / Ext. buck LP4X VDRAM(VDDQ)
I2C address: 0x50 (Write:0xA0, Read:0xA1)

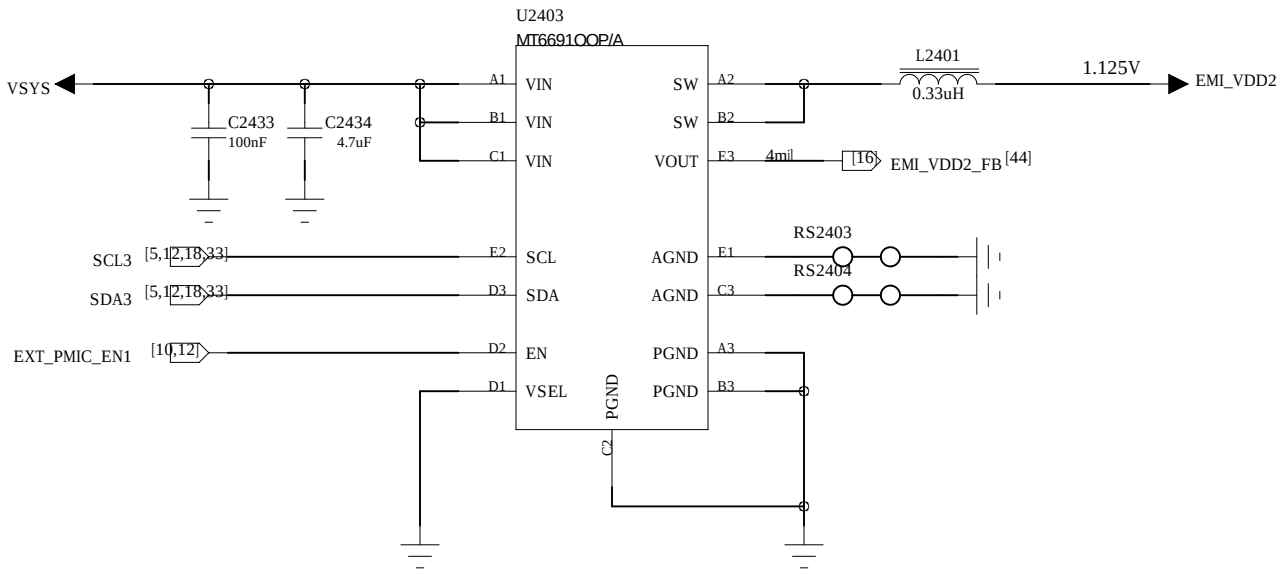
Suggest trace width > 25 mil



Ext. buck LP4/4X VDD2 1.125V

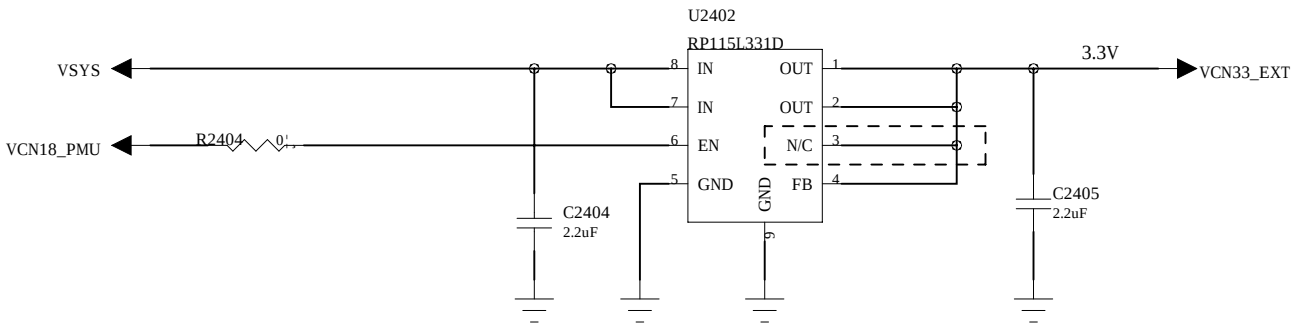
MT6691OOP/A / Ext. buck LP4X VDRAM (VDD2)
I2C address: 0X57 (Write:0xAE, Read:0xAF)

Suggest trace width > 50 mil



LDO for VCN33

Suggest trace width > 25 mil



Schematic design notice of "28_POWER_ThirdParty-Power"

Note 28-1: VA12 Layout placement please close to AP

Note 28-2: VS2 Buck Layout placement please close to PMIC MT6357

Note 28-3: VCN33 LDO Layout placement please close to MT6631

Note 28-4: MT6691ZXP/A and MT6691OOP/A Buck Layout Placement please close to LPDDR4X

Note 28-5: U2810 LDO Layout Placement Please close to LPDDR4X VDD1 power ball

Note 28-6: For EML_VDD2_FB and EML_VDDQ_FB, please follow MMD rule

Schematic design notice of "28_POWER_ThirdParty-Power"

Note 28-1: VA12 Layout placement please close to AP

Note 28-2: VS2 Buck Layout placement please close to PMIC MT6357

Note 28-3: VCN33 LDO Layout placement please close to MT6631

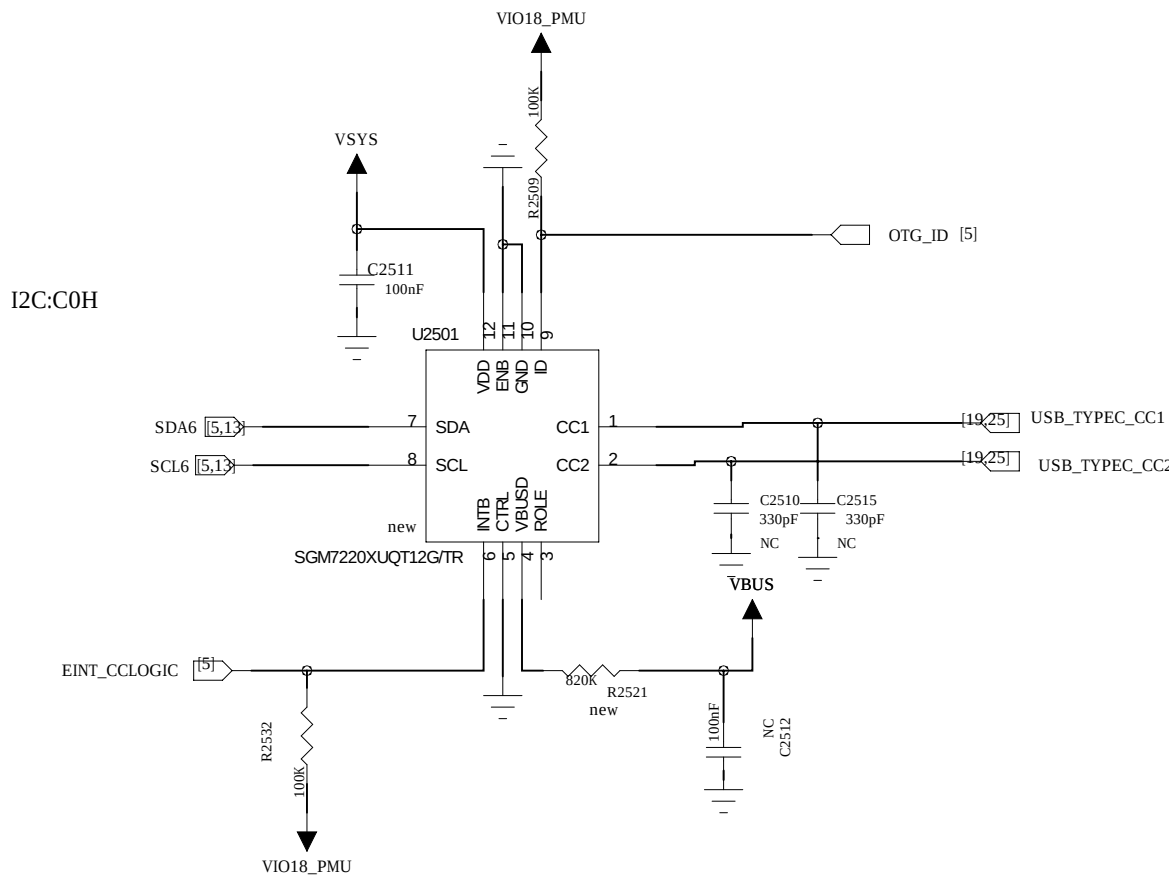
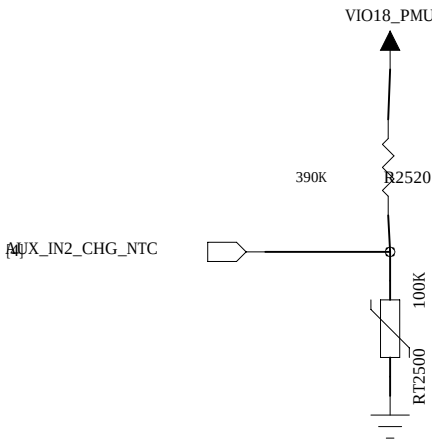
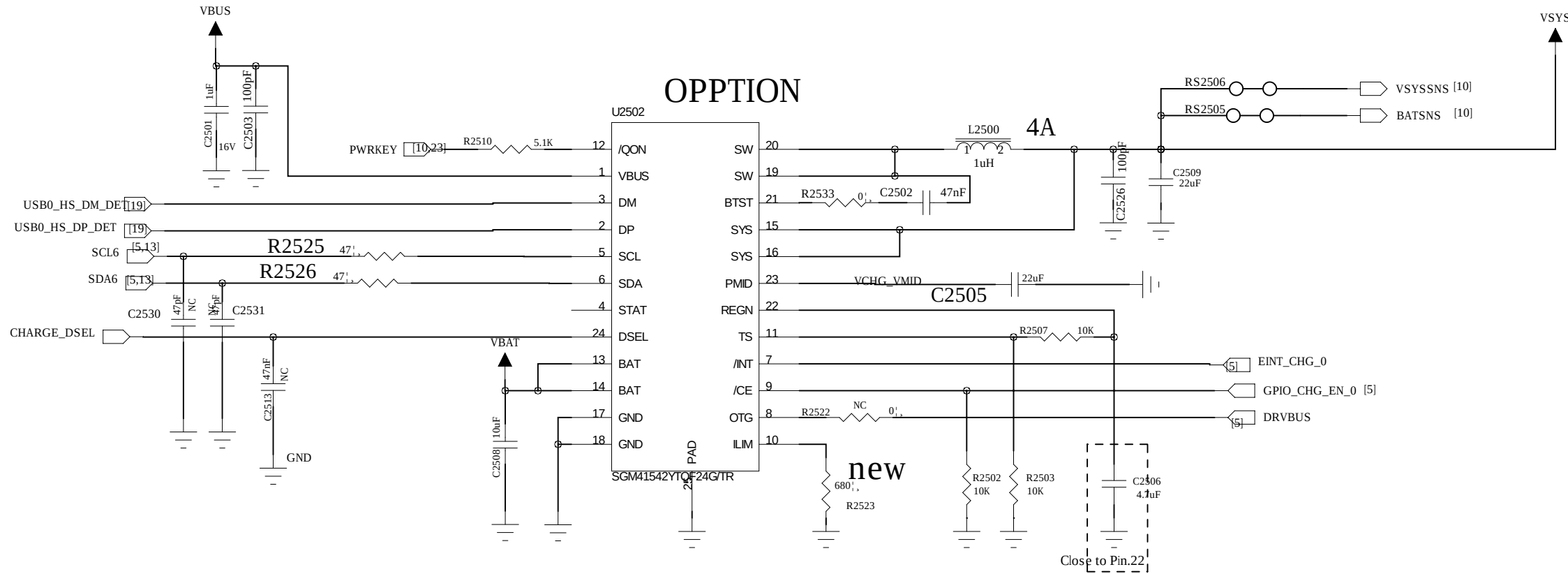
DRAWN:	Marshall	DATED:	2018-09-22A
CHECKED:	<Checked By>	DATED:	<Checked Date>
QUALITY CONTROL:	<QC By>	DATED:	<QC Date>
RELEASED:	<Released By>	DATED:	<Release Date>

COMPANY:		Tinno	
TITLE:		MT6765	
CODE:	SIZE:	DRAWING NO:	REV:
P410	D	P410	V1.0
SCALE: <Scale>		SHEET: 42	33

CHARGE 9V*2A

I2C ADDR:0x6B

L2501,L2502,R2501,R2504 CLOSE TO Connector!!!



REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE

COMPANY: Tinno			
TITLE: MT6765			
DRAWN: Marshall	DATED: 2018-09-22A		
CHECKED: <Checked By>	DATED: <Checked Date>		
QUALITY CONTROL: <QC By>	DATED: <QC Date>		
RELEASED: <Released By>	DATED: <Release Date>		
CODE: P410		SIZE: D	DRAWING NO: P410
			REV: V1.0
SCALE: <Scale>			SHEET: 33

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REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

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A

		COMPANY: Tinnor			
		TITLE: MT6765			
DRAWN: Marshall	DATED: 2018-09-22A				
CHECKED: <Checked By>	DATED: <Checked Date>	CODE: P410	SIZE: D	DRAWING NO: P410	REV: V1.0
QUALITY CONTROL: <QC By>	DATED: <QC Date>				
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>			SHEET: 44 33

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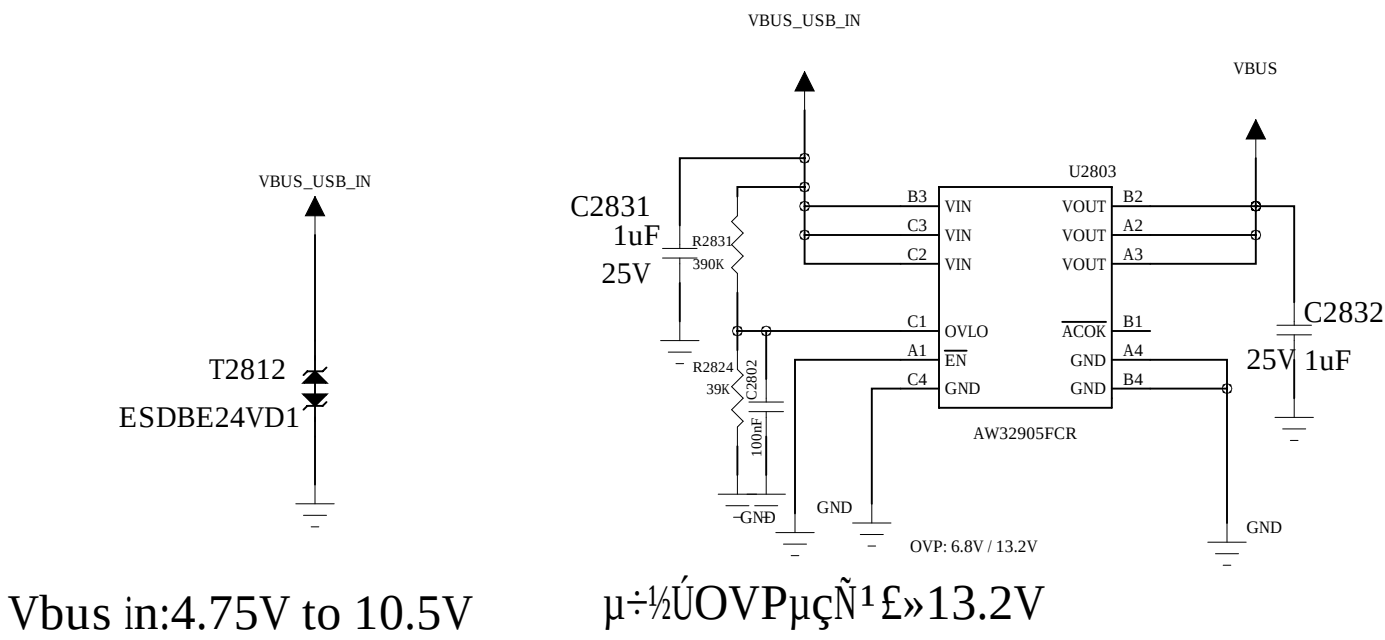
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1

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



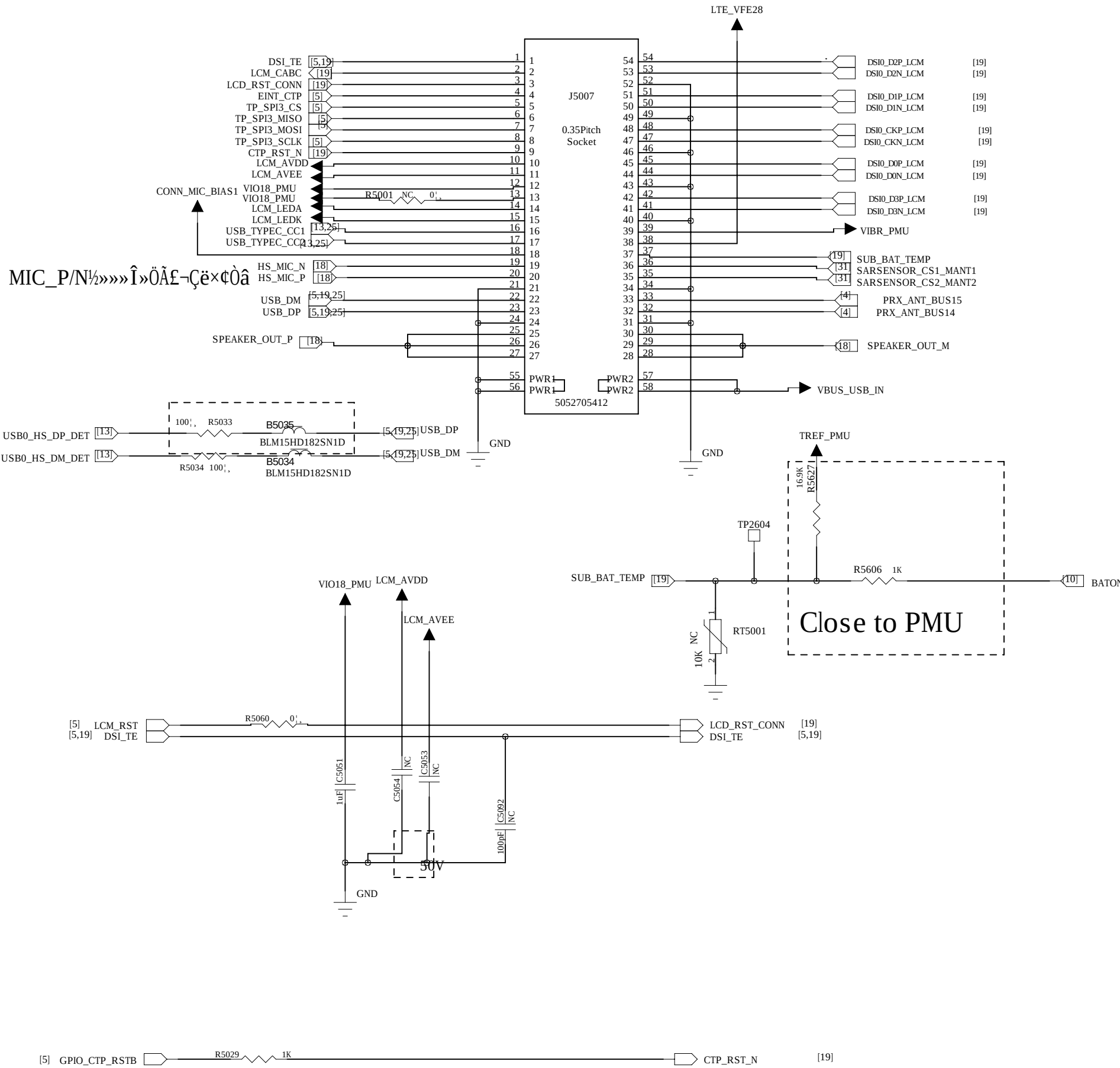
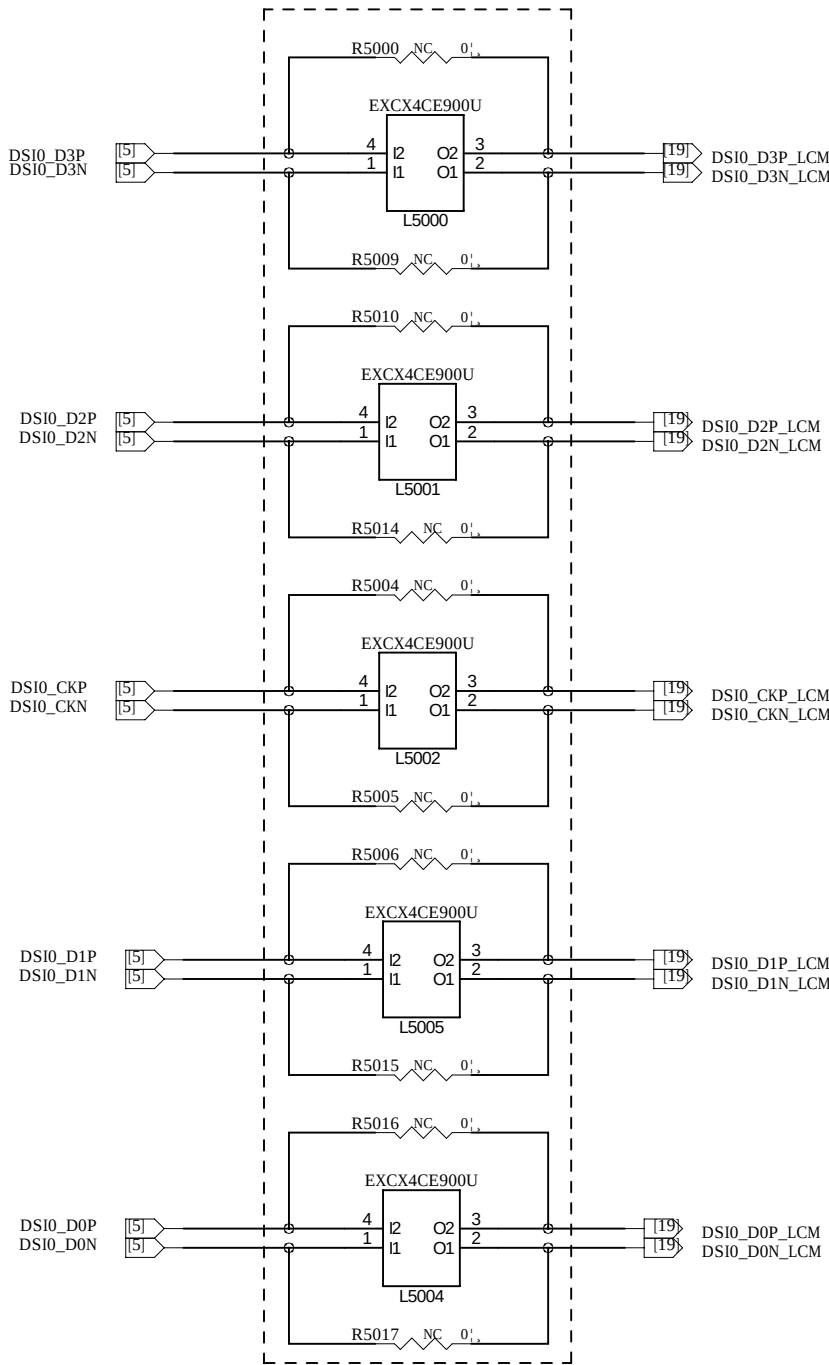
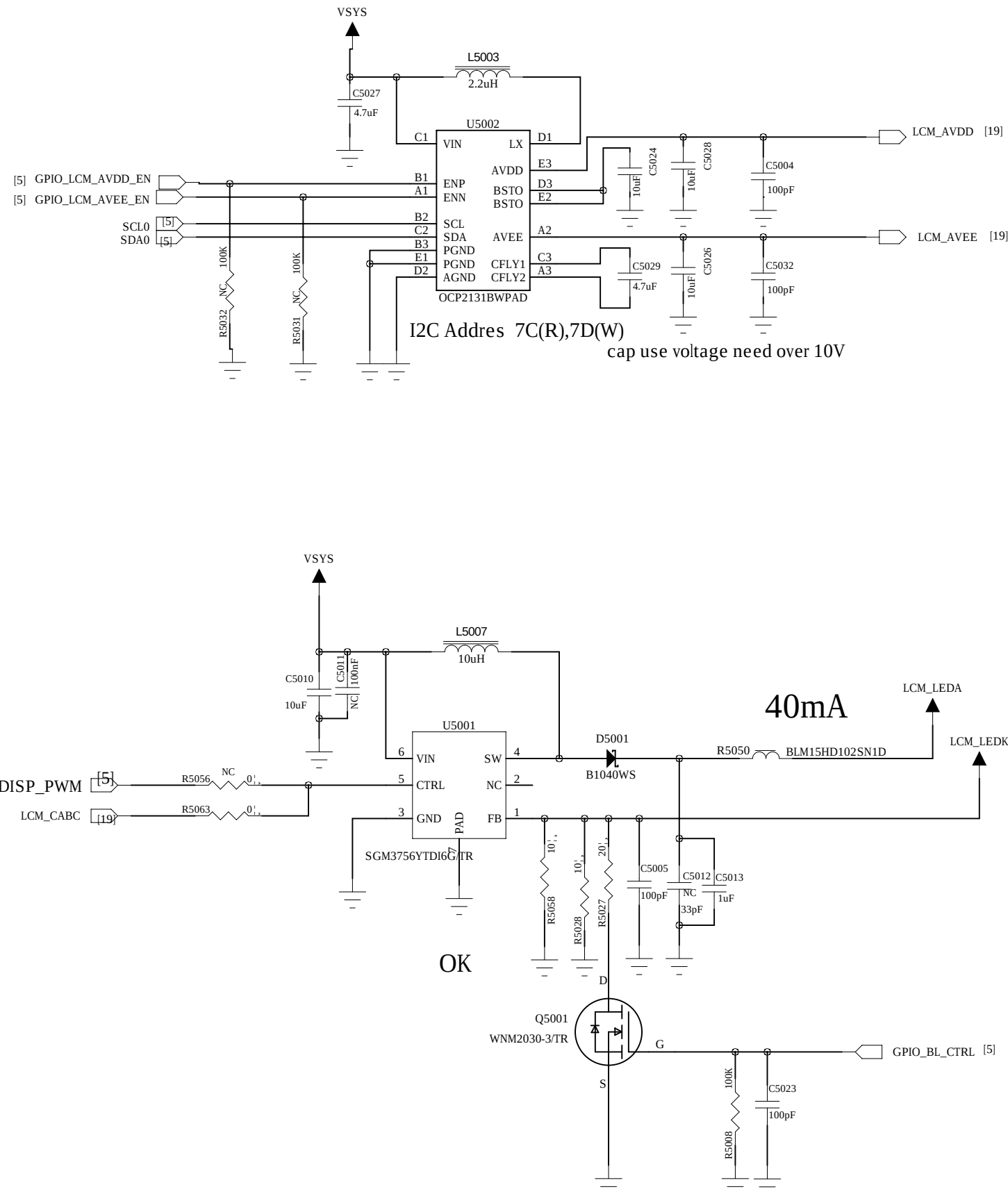
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TITLE:		MT6765			
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P410	D	P410		V1.0	
SCALE: <Scale>			SHEET: 45 33		

DRAWN:	Marshall	DATED:	2018-09-22A
CHECKED:	<Checked By>	DATED:	<Checked Date>
QUALITY CONTROL:	<QC By>	DATED:	<QC Date>
RELEASED:	<Released By>	DATED:	<Release Date>

COBY-P352CE

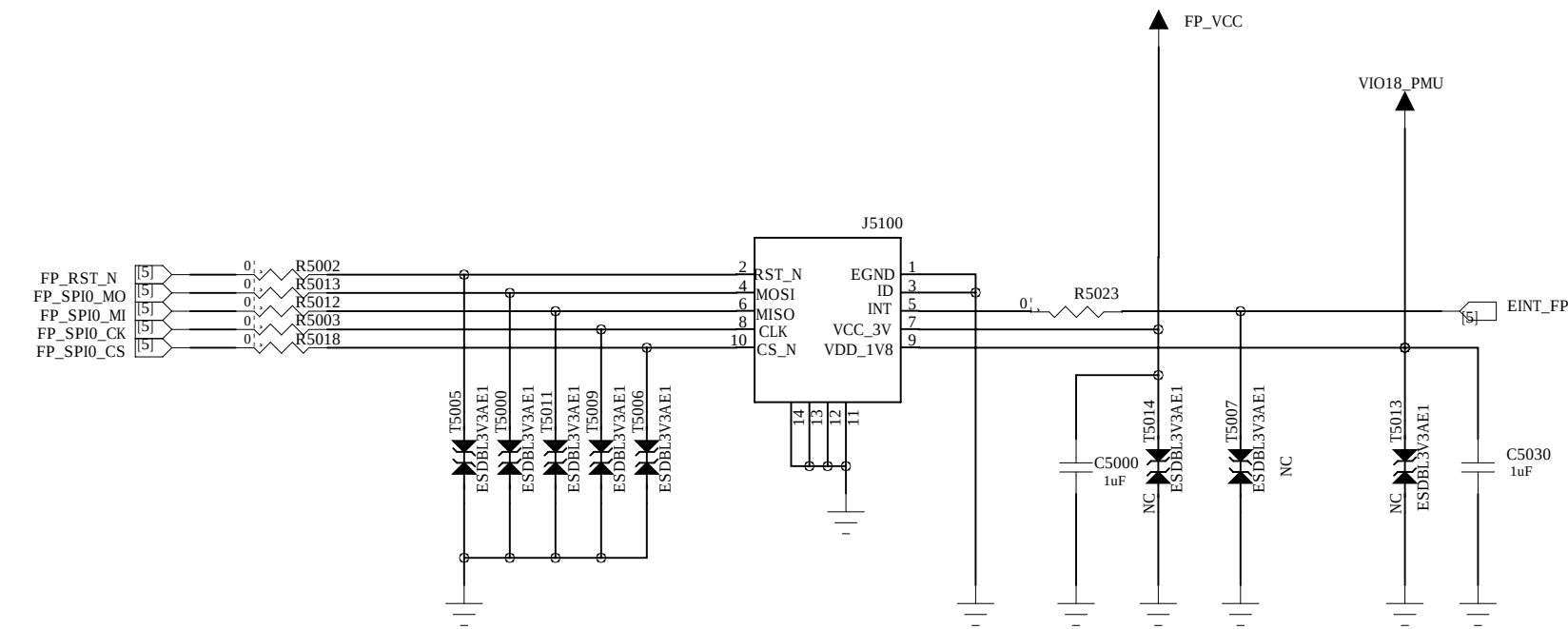
LCD Connector

REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE

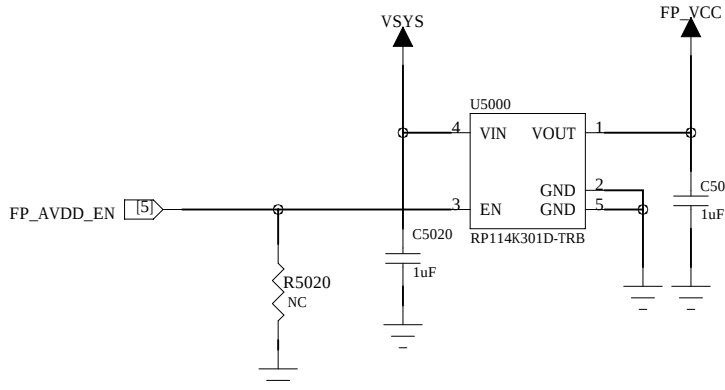


COBY-P352CE

Fingerprint Connector



1	GND
2	RST
3	ID(GND)
4	MOSI
5	INT
6	MISO
7	VDD2V8
8	CLK
9	DOVDD1V8
10	CS
11	GND
12	GND
13	GND
14	GND



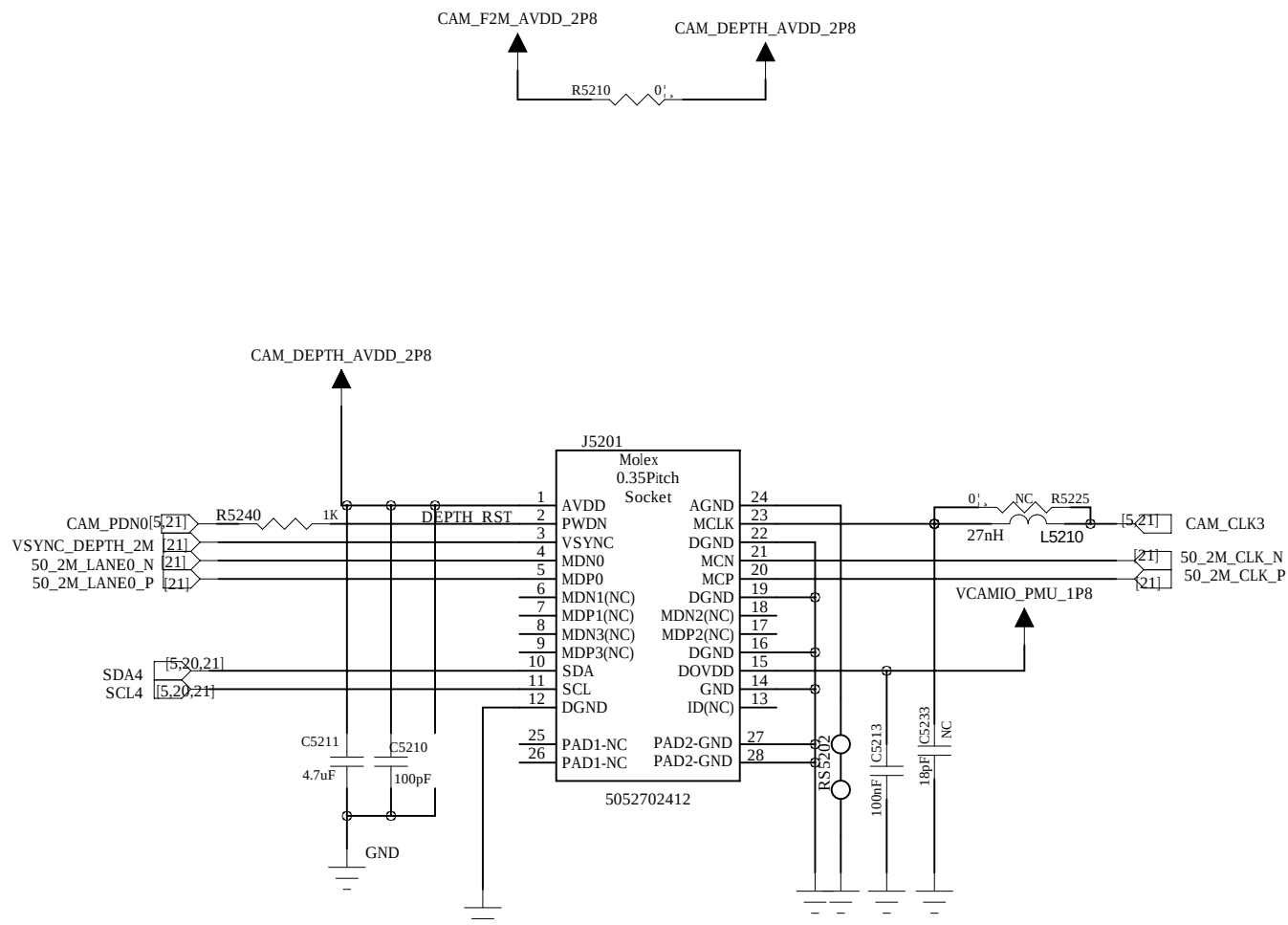
COMPANY: Tinnno			
TITLE: MT6765			
DRAWN: Marshall	DATED: 2018-09-22A	CODE: P410	
CHECKED: <Checked By>	DATED: <Checked Date>	SIZE: D	DRAWING NO: P410
QUALITY CONTROL: <QC By>	DATED: <QC Date>	REV: V1.0	
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>	
SHEET 40			33

Camera LDO

Main-CAMERA

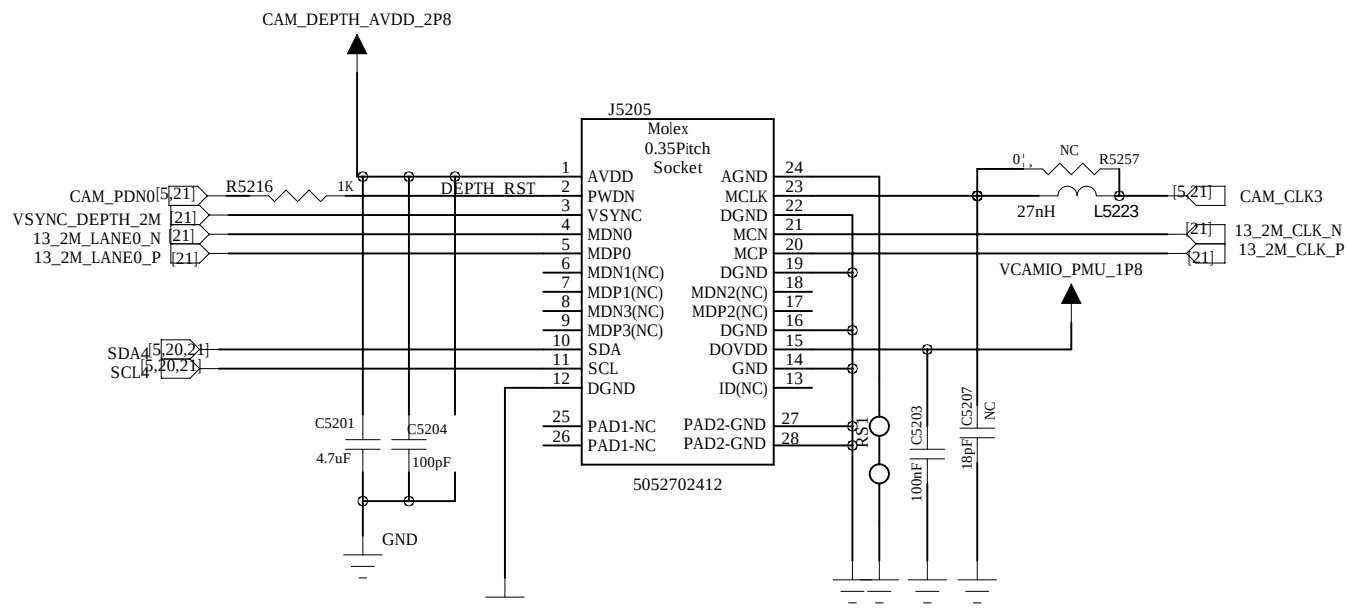
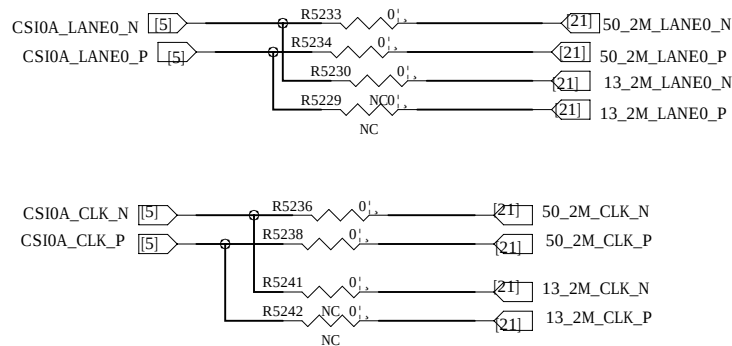
50M/13M

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R12,R17,R22,R25 Close to J5201 !!!!

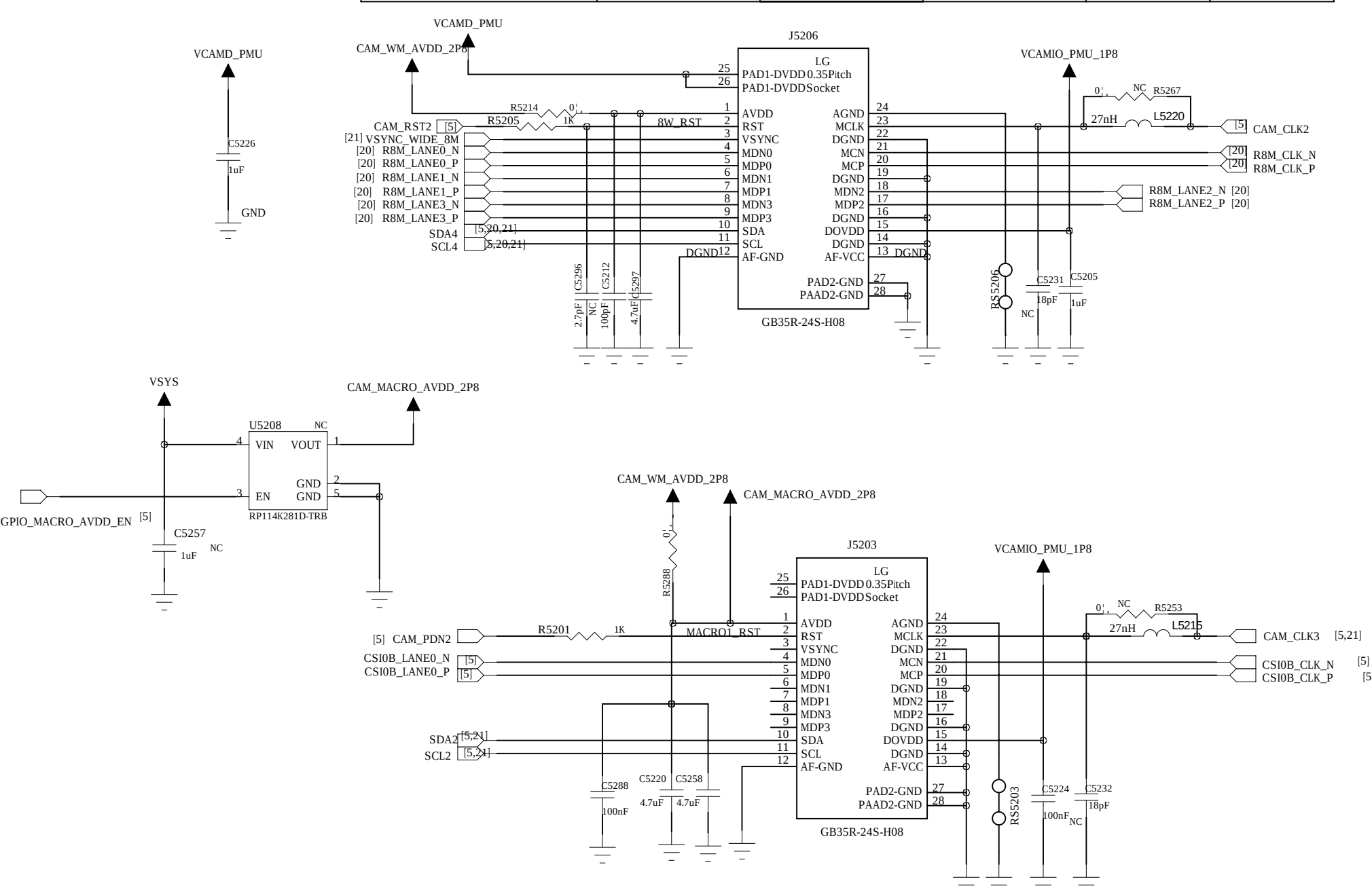
2M FF-Depth Side-50M



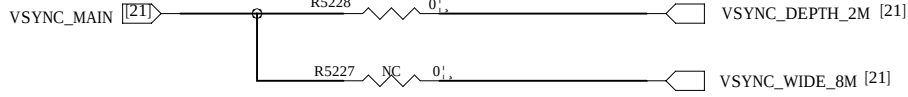
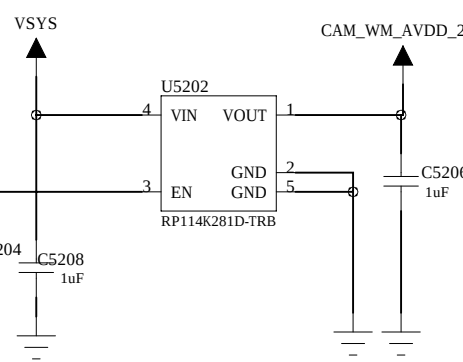
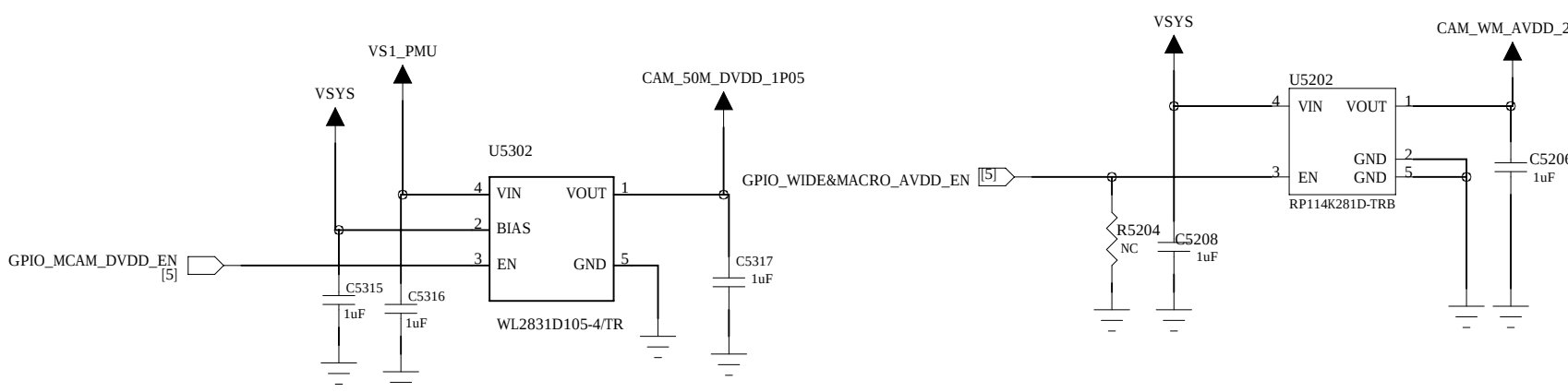
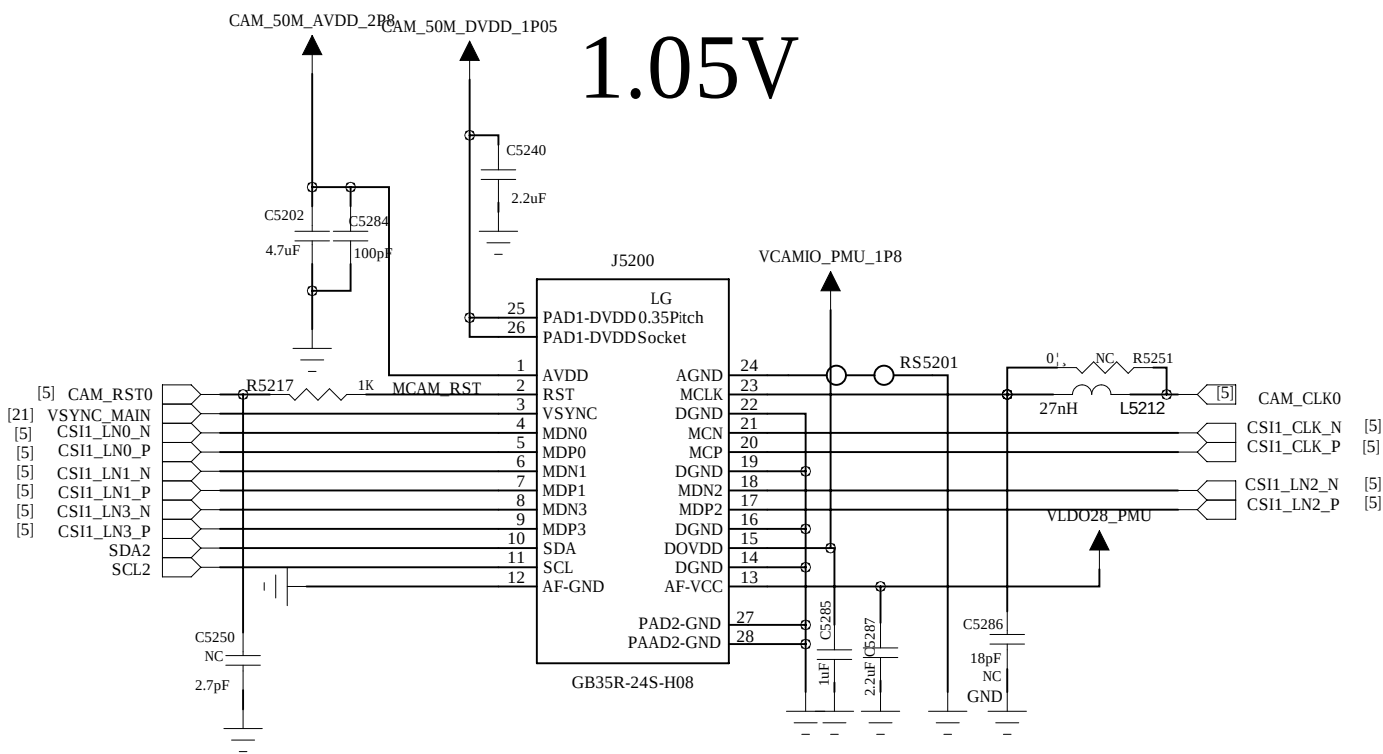
2M FF-Depth Side-13M

8M: WIDE

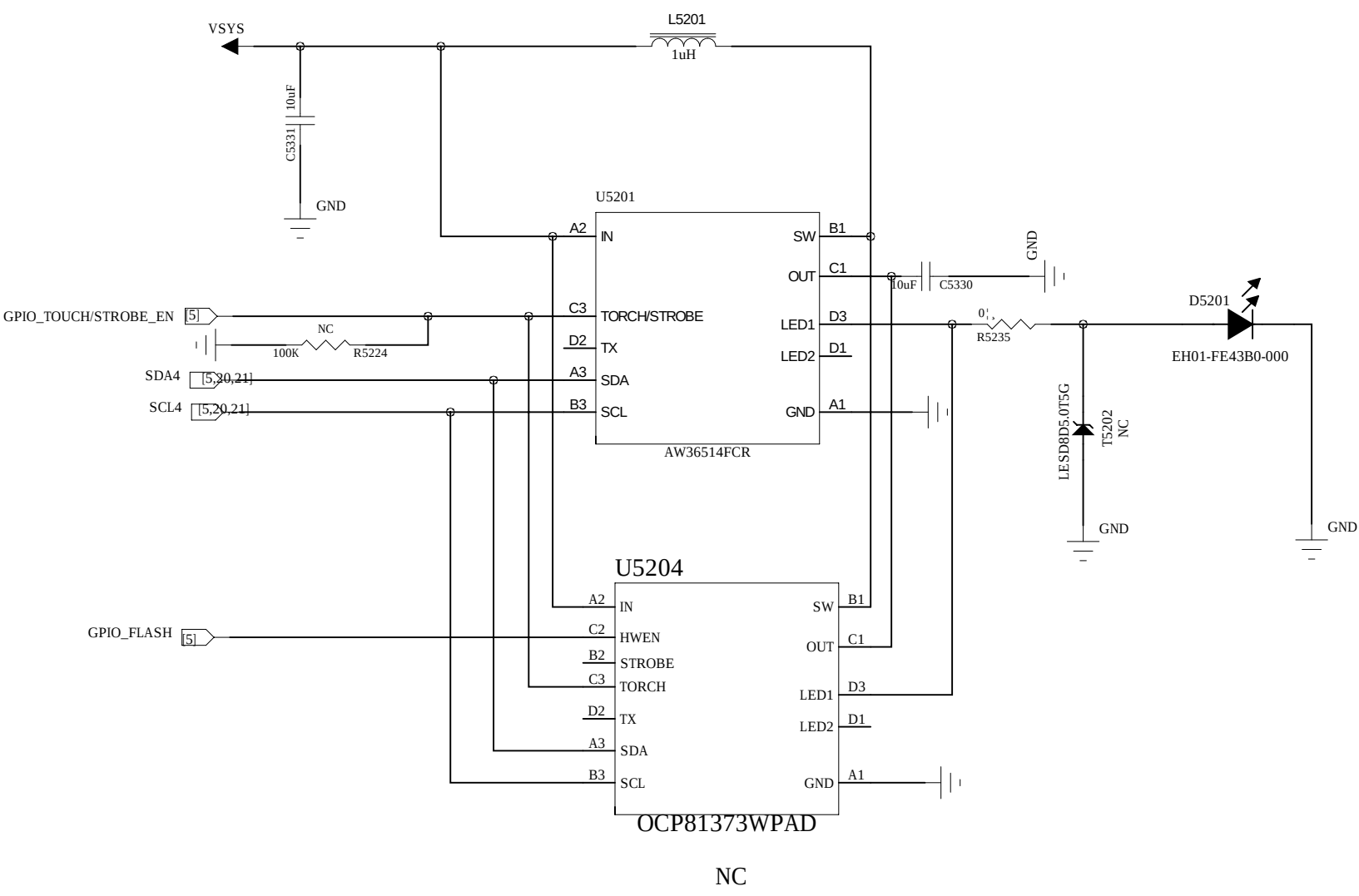
	AVDD	DVDD	DOVDD	ID	I2C
S5K4H7YX03	2.7~2.9 V(2.8V)	1.14~1.26V(1.2V)	1.7~1.9V(1.8V)		0X5A(W)0X5B(R)



2M FF-Macro



Rear CAM Flash LED



COMPANY:		Tinno	
TITLE:		MT6765	
CAMERA			
CODE:	SIZE:	DRAWING NO:	REV:
P410	D	P410	V1.0
SCALE: <Scale>		SHEET: ad	33

	AVDD	DVDD	DOVDD	I2C
GC5035	2.7V-3.0V	1.15V-1.3V	1.7V-3.0V	W:0x7E R:0x7F

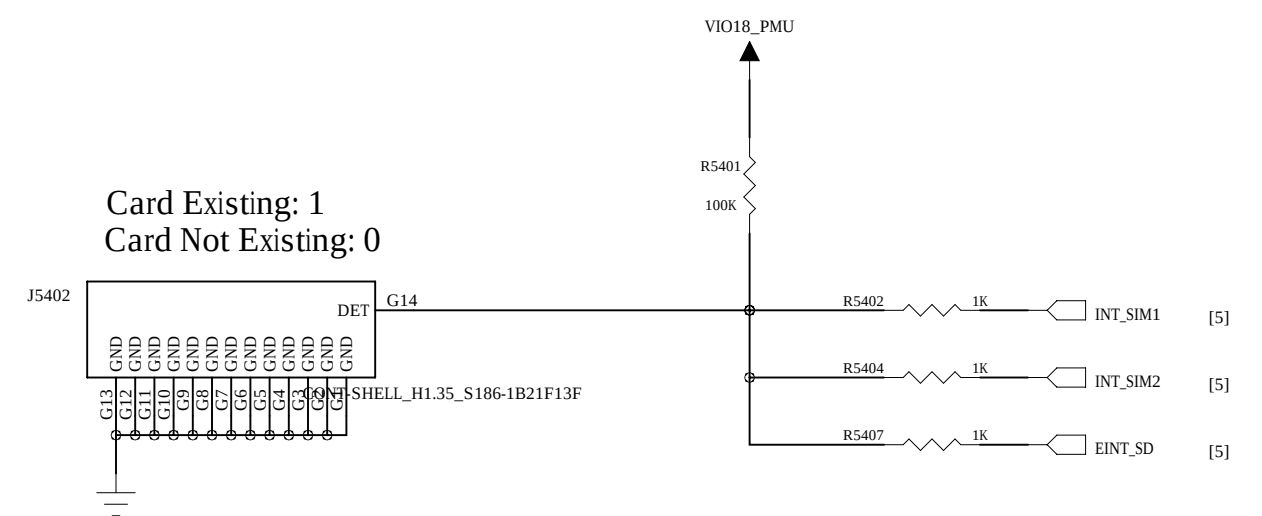
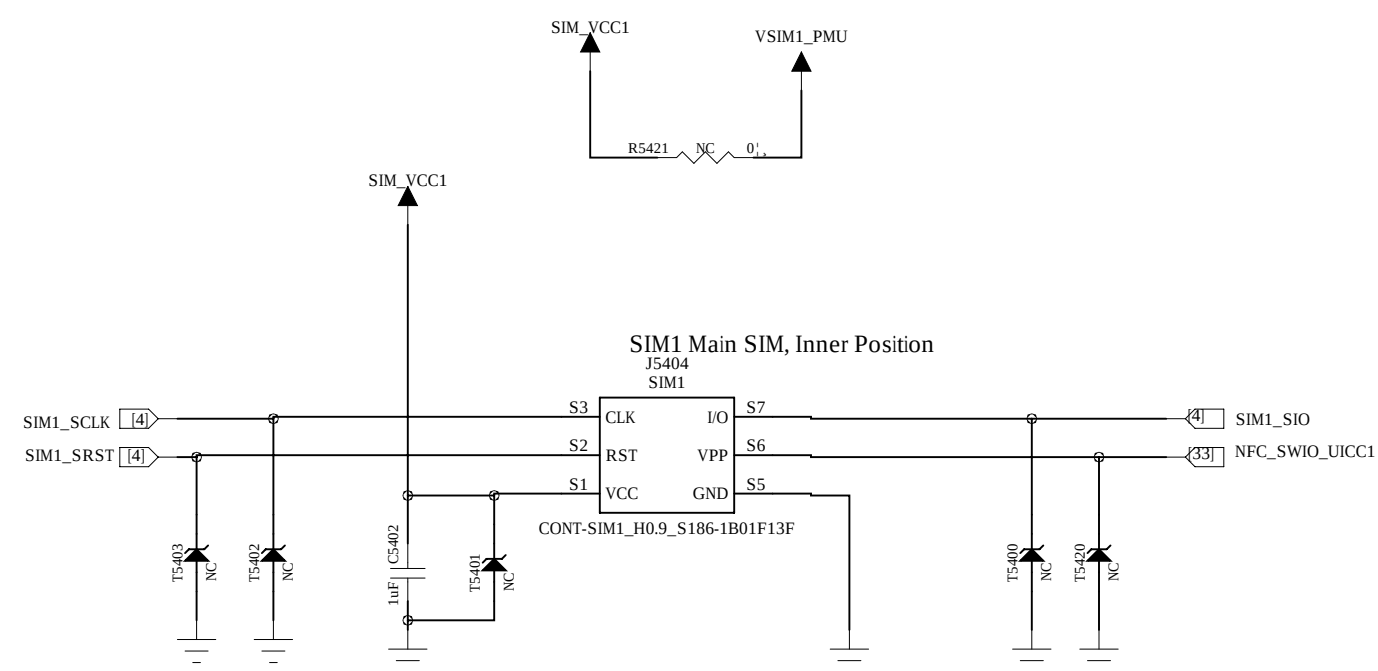
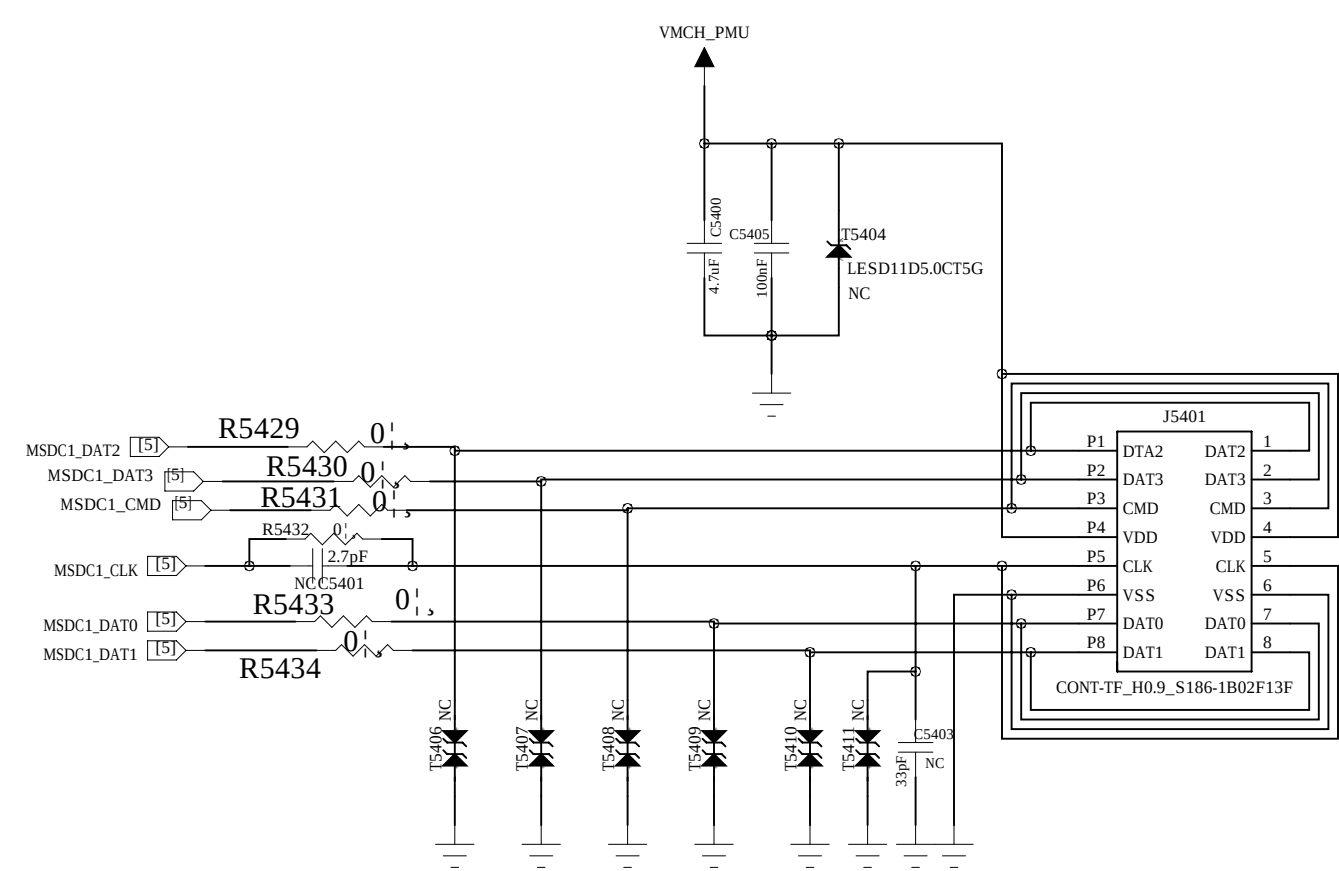
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

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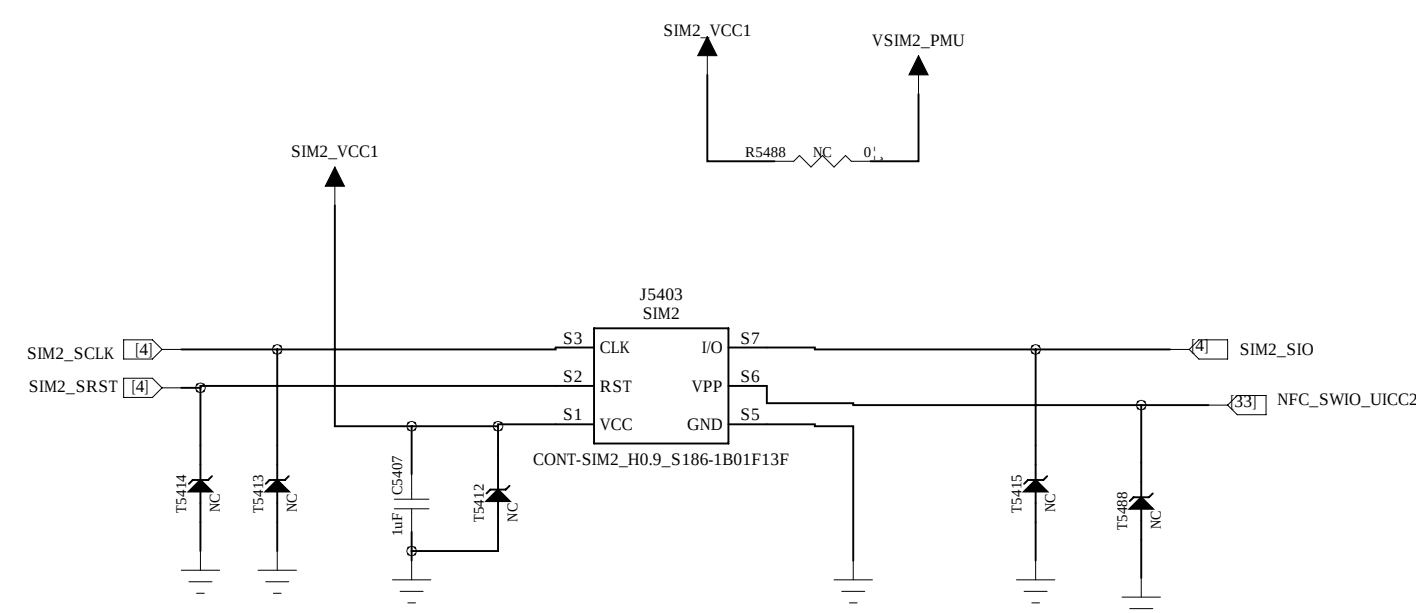
C



E

B

SIM Card 2



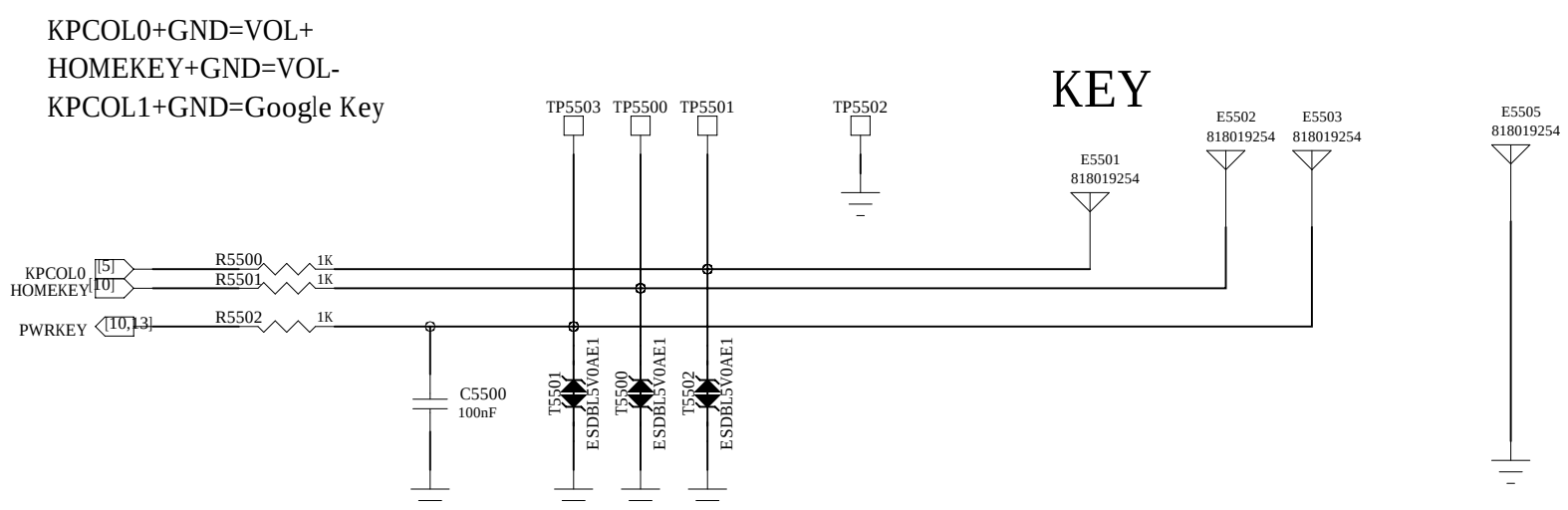
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A

		COMPANY: Tino			
		TITLE: MT6765			
DRAWN: Marshall	DATED: 2018-09-22A				
CHECKED: <Checked By>	DATED: <Checked Date>	CODE: P410	SIZE: D	DRAWING NO: P410	REV: V1.0
QUALITY CONTROL: <QC By>	DATED: <QC Date>				
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>			SHEET: 22 33

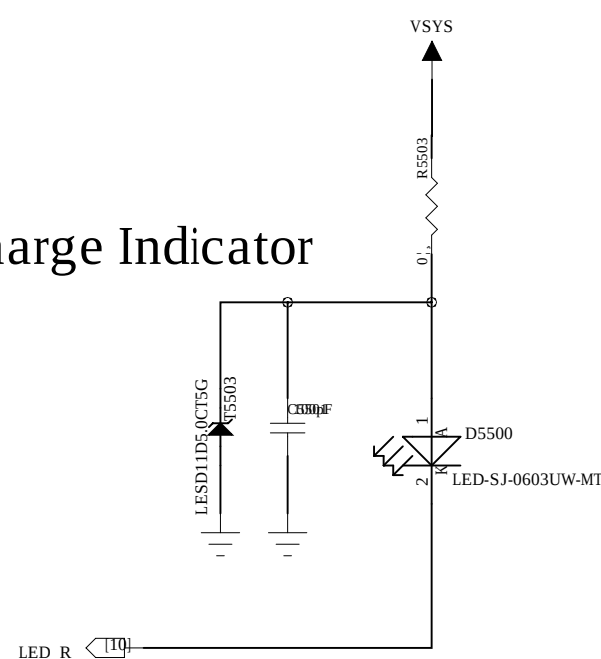
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

MP Test point



COBY-P720AE

Charge Indicator



DRAWN: Marshall CHECKED: <Checked By> QUALITY CONTROL: <QC By> RELEASED: <Released By>		DATED: 2018-09-22A		COMPANY: Tino			
		TITLE: MT6765					
		CODE: P410	SIZE: D	DRAWING NO: P410		REV: V1.0	
		SCALE: <Scale>		SHEET: 08 33			

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

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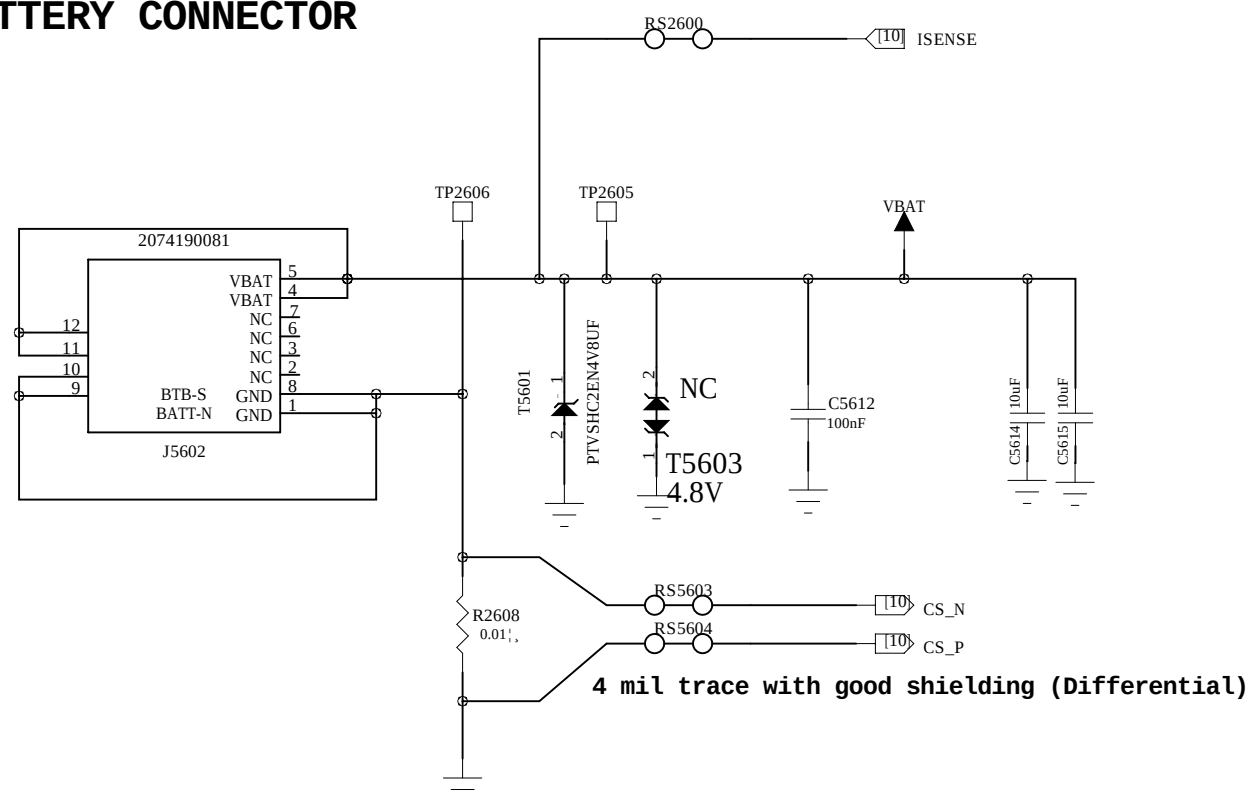
D

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BATTERY CONNECTOR



DRAWN: Marshall	DATED: 2018-09-22A
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

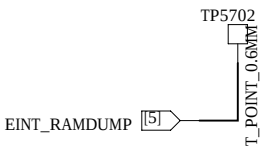
COMPANY:		Tinno	
TITLE: MT6765			
CODE: P410	SIZE: D	DRAWING NO: P410	REV: V1.0
SCALE: <Scale>		SHEET: a4	33

Debug Test Point

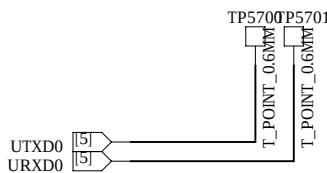
Fixture Test Point

REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE

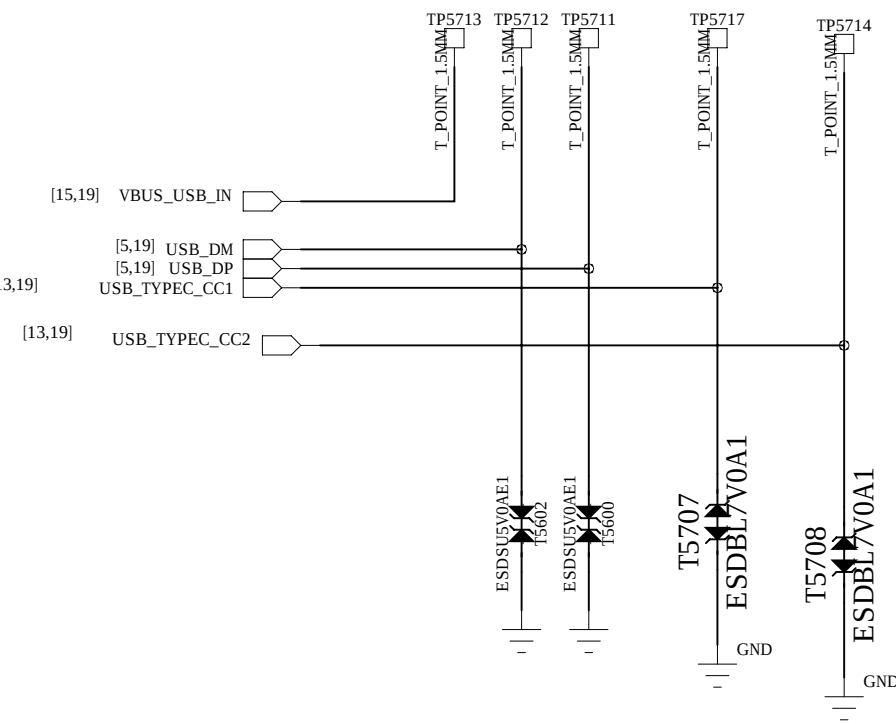
RAM dump



UART



USB

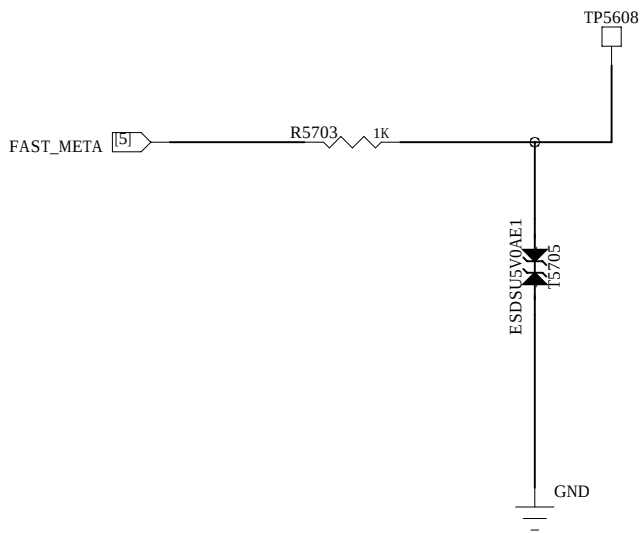
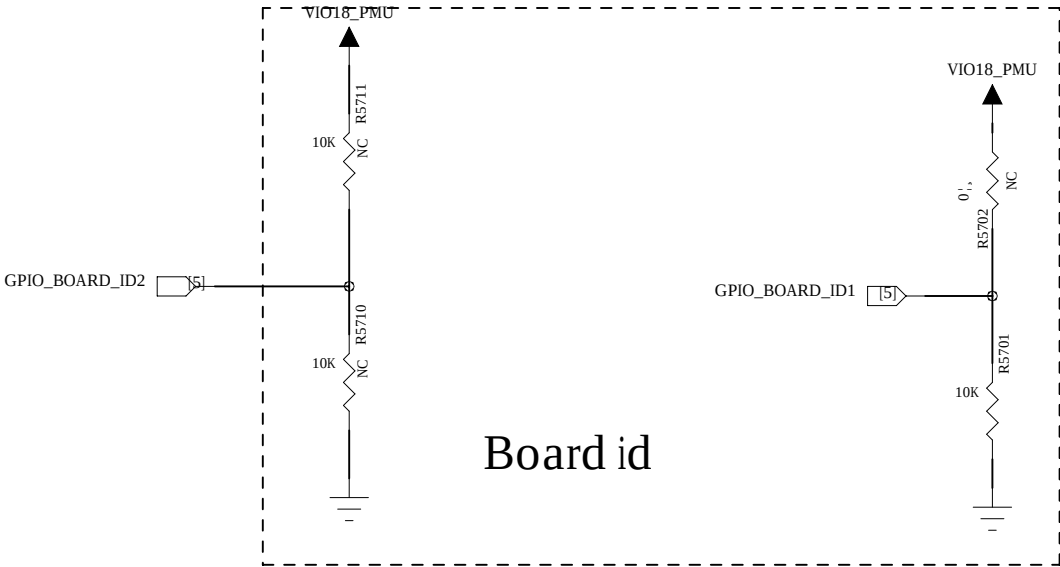


Batery
Refer to Page 26

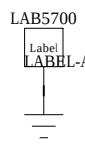
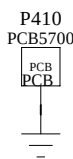
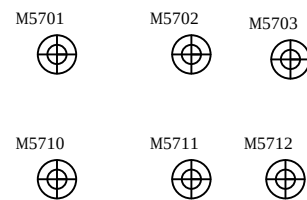
CTP
Refer to Page 56

Key
Refer to Page 55

Fast META

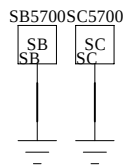


Mark

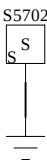


shielding can

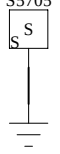
BB



PDN



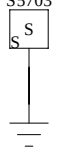
RF



4G PA

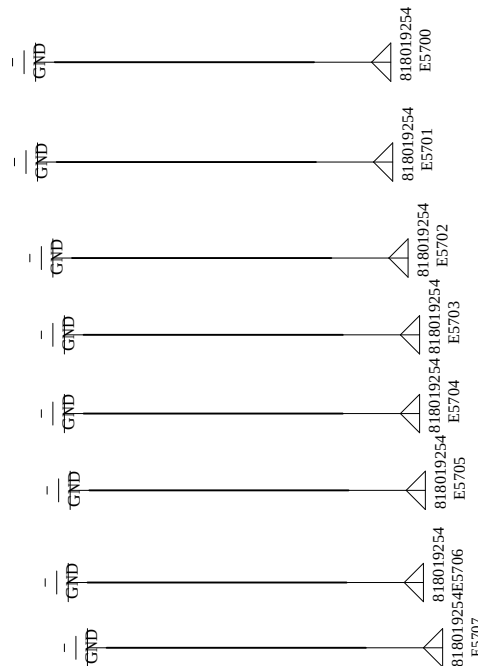


WCN



Metal Grounding

Top

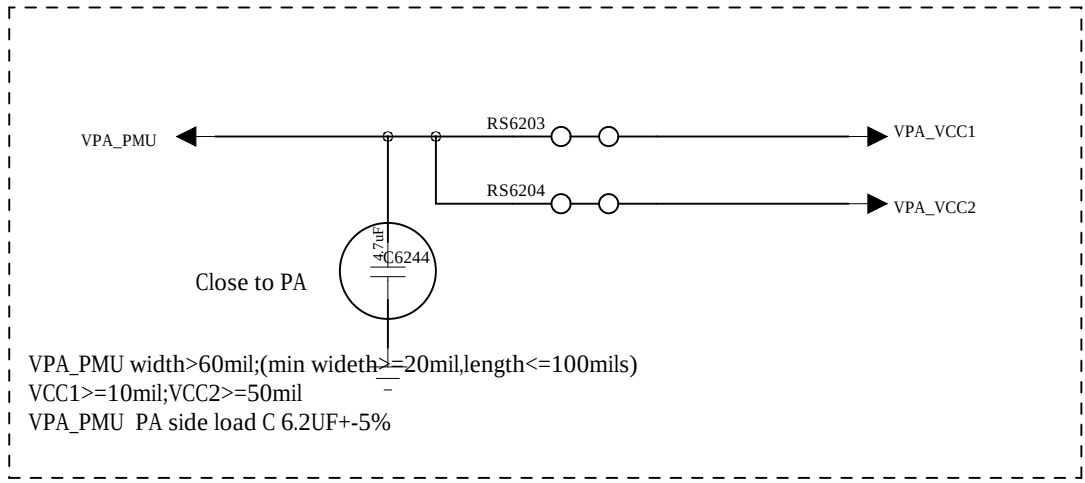


Bottom

COMPANY: Tinno			
TITLE: MT6765			
DRAWN: Marshall	DATED: 2018-09-22A		
CHECKED: <Checked By>	DATED: <Checked Date>	CODE: P410	SIZE: D
QUALITY CONTROL: <QC By>	DATED: <QC Date>	DRAWING NO: P410	REV: V1.0
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>	SHEET: 35 33

REVISION RECORD			
LTR	ECD NO.	APPROVED:	DATE:

CO-layer	C6243	C6247
B4	NC	33pF
B41	33pF	NC



VCC1

Band	RX
B40(pin37)	TRX1(pin41)
B41(pin39)	TRX2(42) pin33,pin35

3/4G_PAIN_HB

3/4G_PAIN_MB

3/4G_PAIN_LB

4C PA input need LPF for harmonic rejection

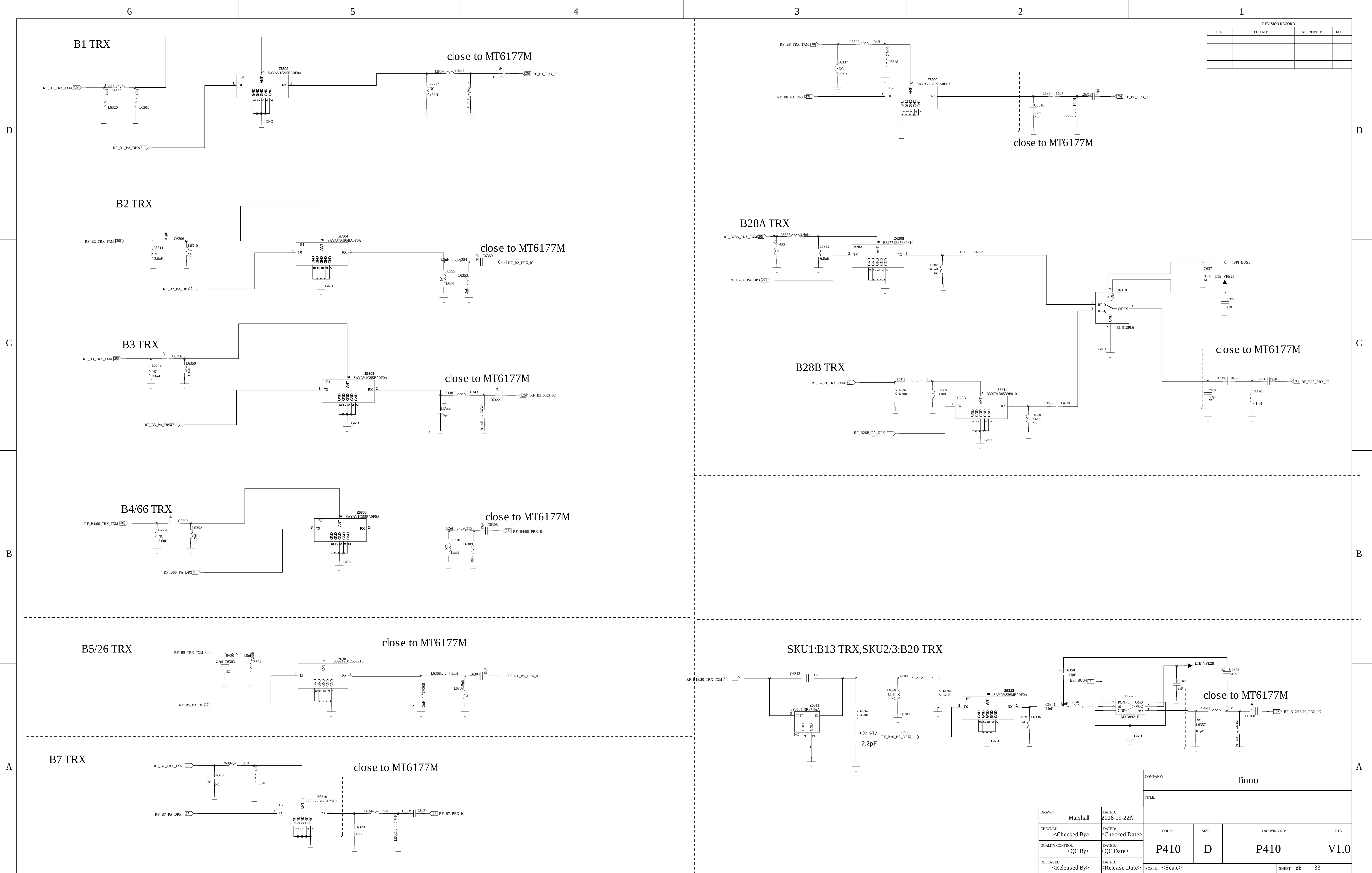
Thermistor / To sense board level temperature

Thermistor have to be placed as close to B7&B1 PA as possible, routing with ground guard.
RT1900 close to PA , and located in the same layer

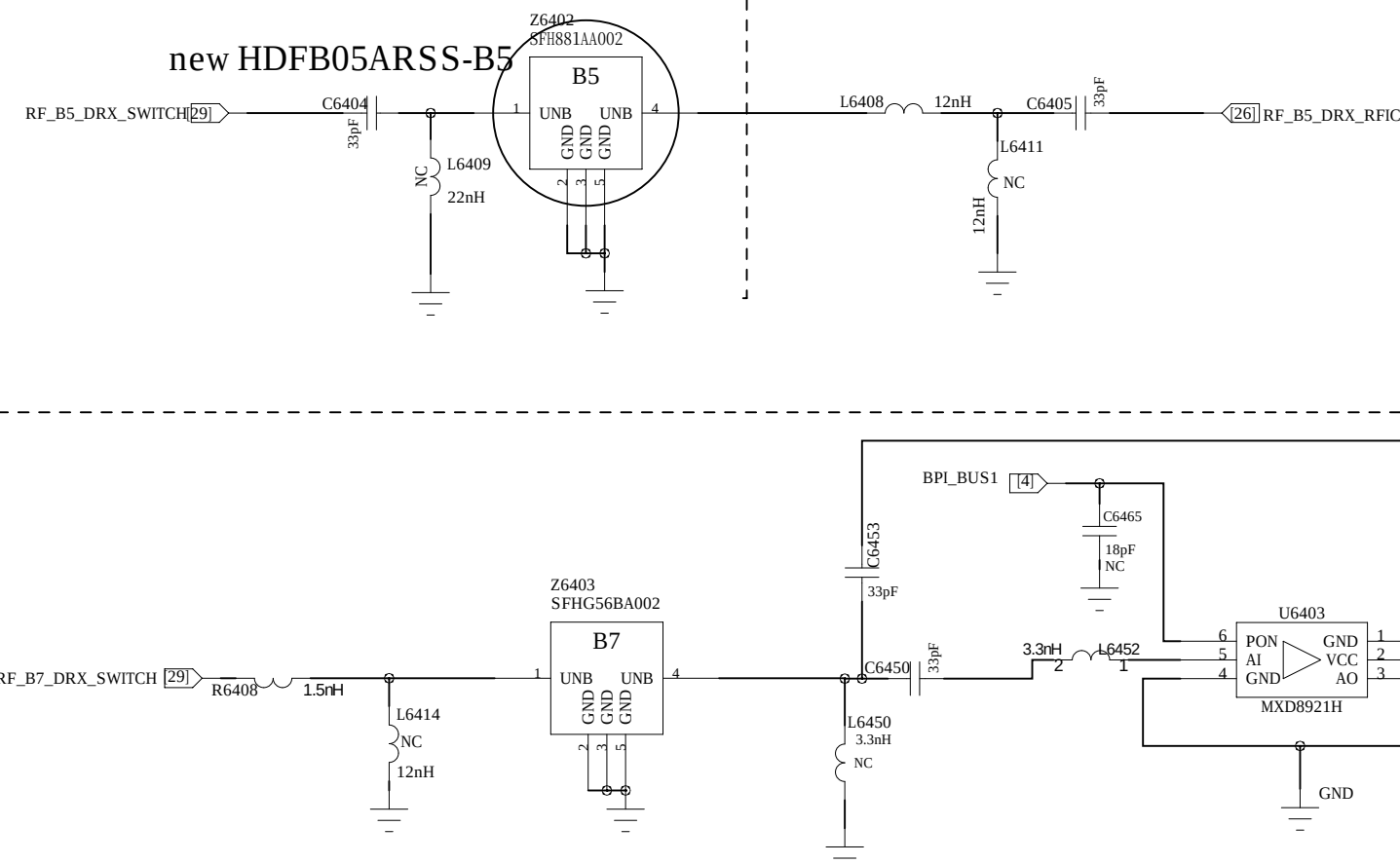
close to MT6177M

DRAWN:	Marshall	DATED:	2018-09-22A
CHECKED:	<Checked By>	DATED:	<Checked Date>
QUALITY CONTROL:	<QC By>	DATED:	<QC Date>
RELEASED:	<Released By>	DATED:	<Release Date>

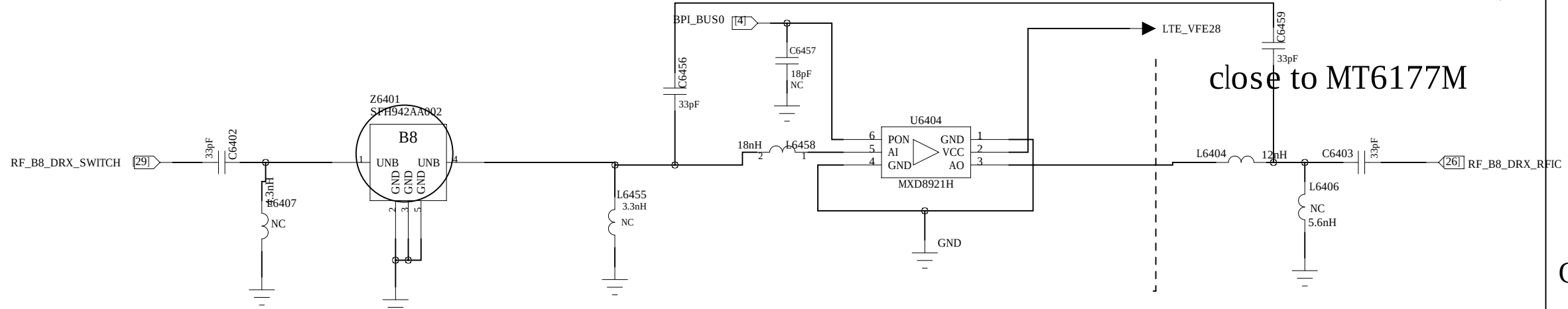
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TITLE: MT6765			
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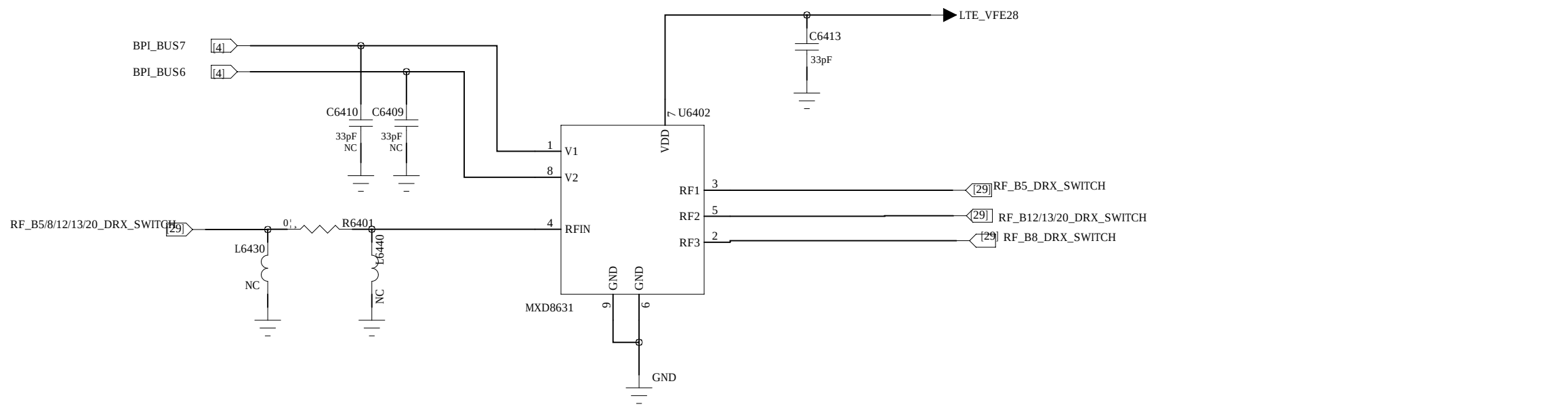
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P410	D	P410	V1.0
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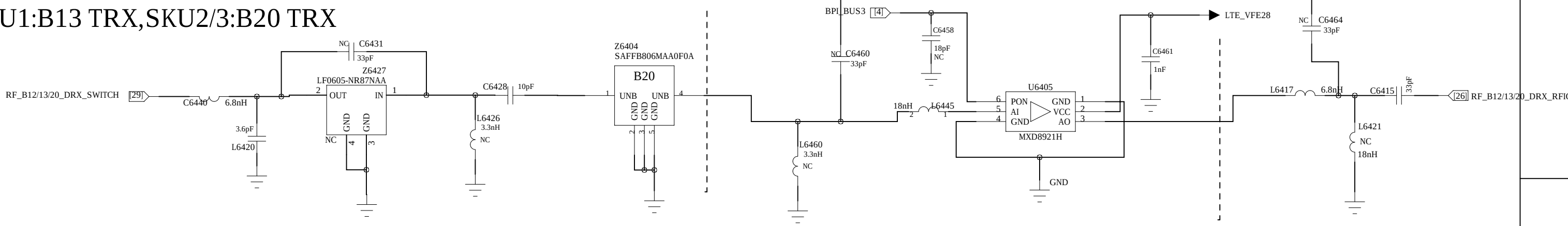
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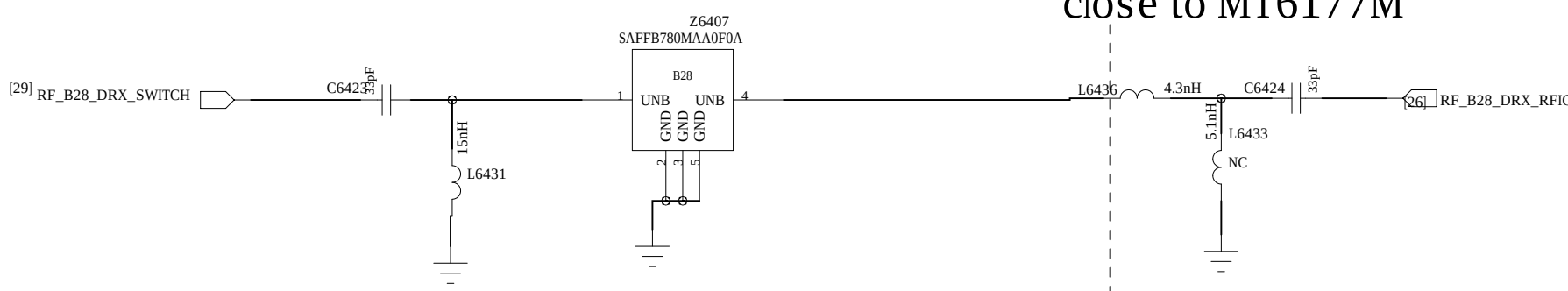
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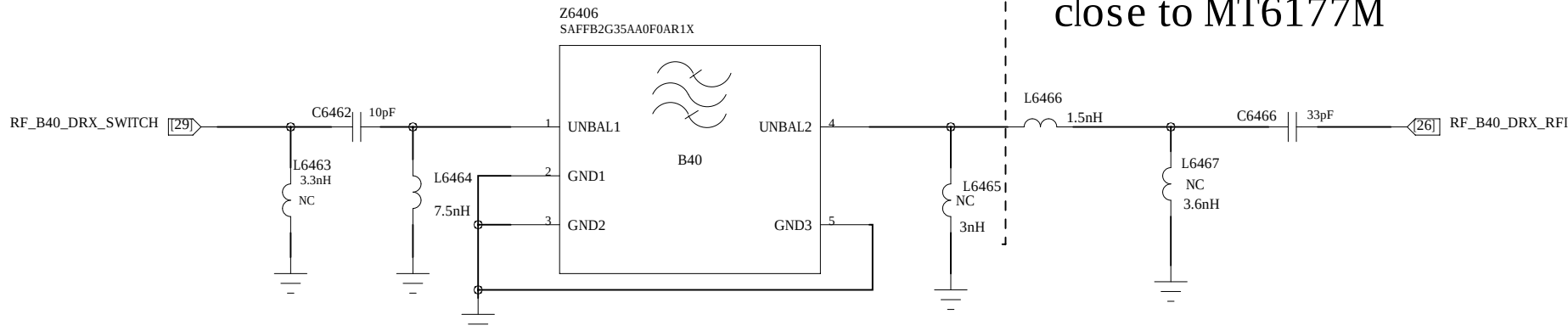
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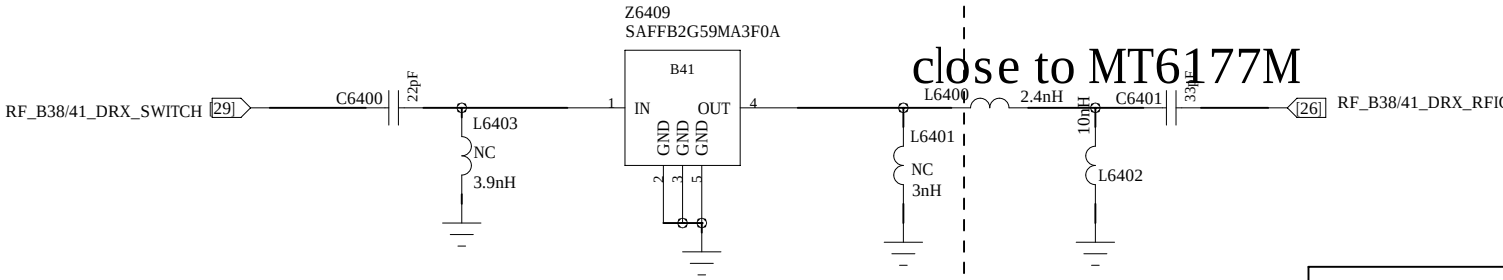
close to MT6177M



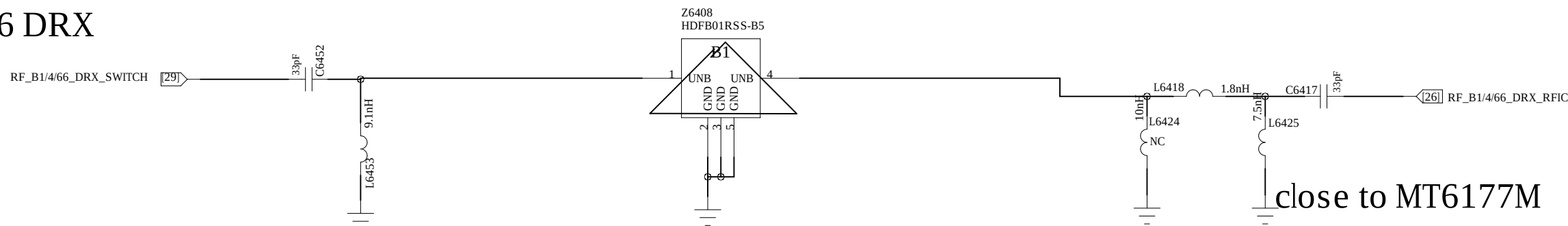
close to MT6177M



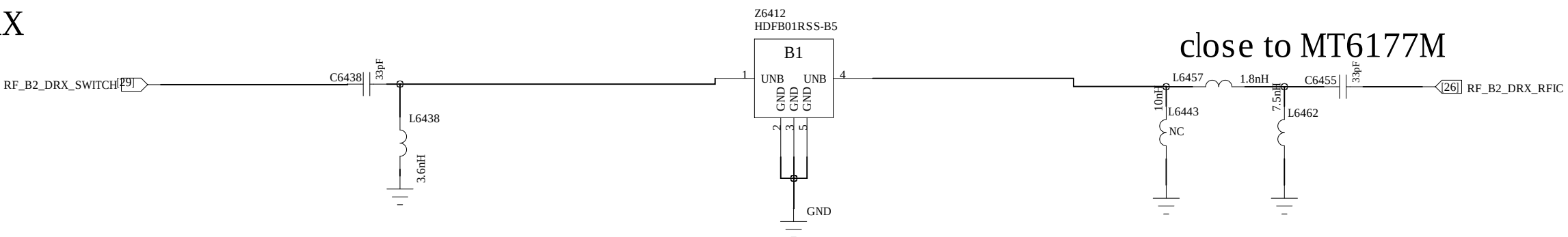
close to MT6177M



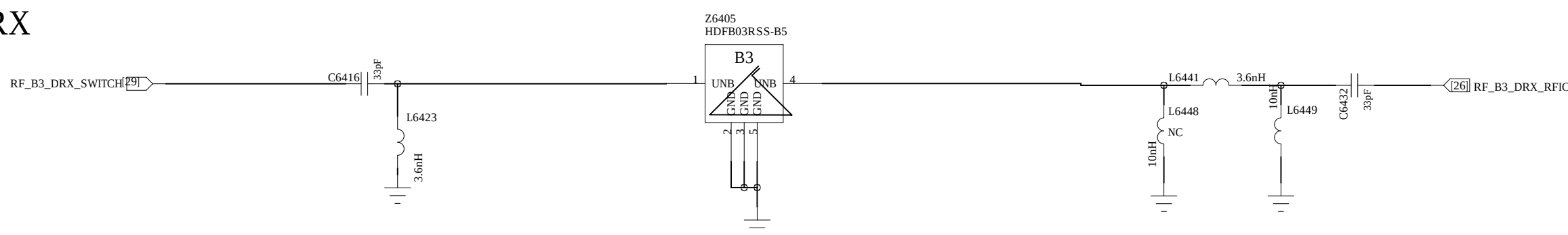
_close to MT6177M



close to MT6177M



RF_B3_DRX_RFIC



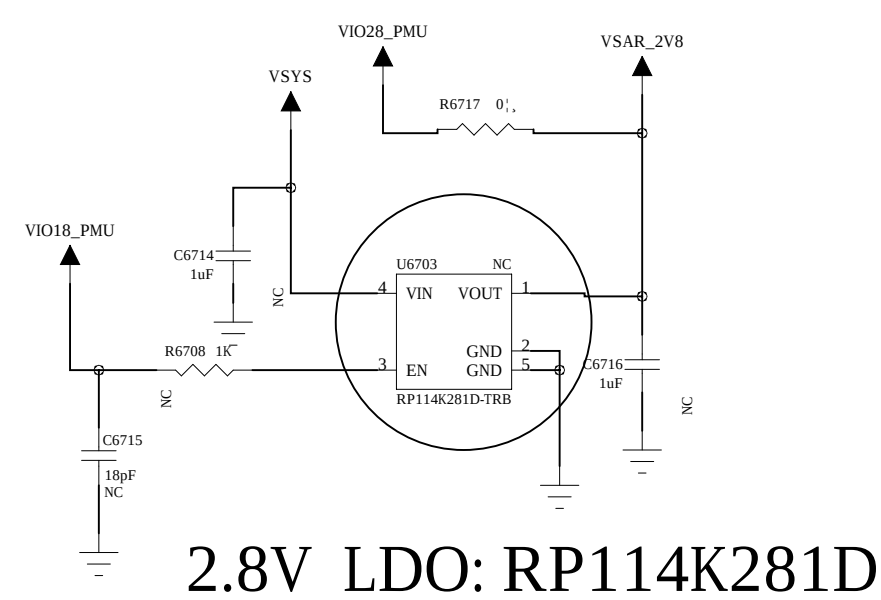
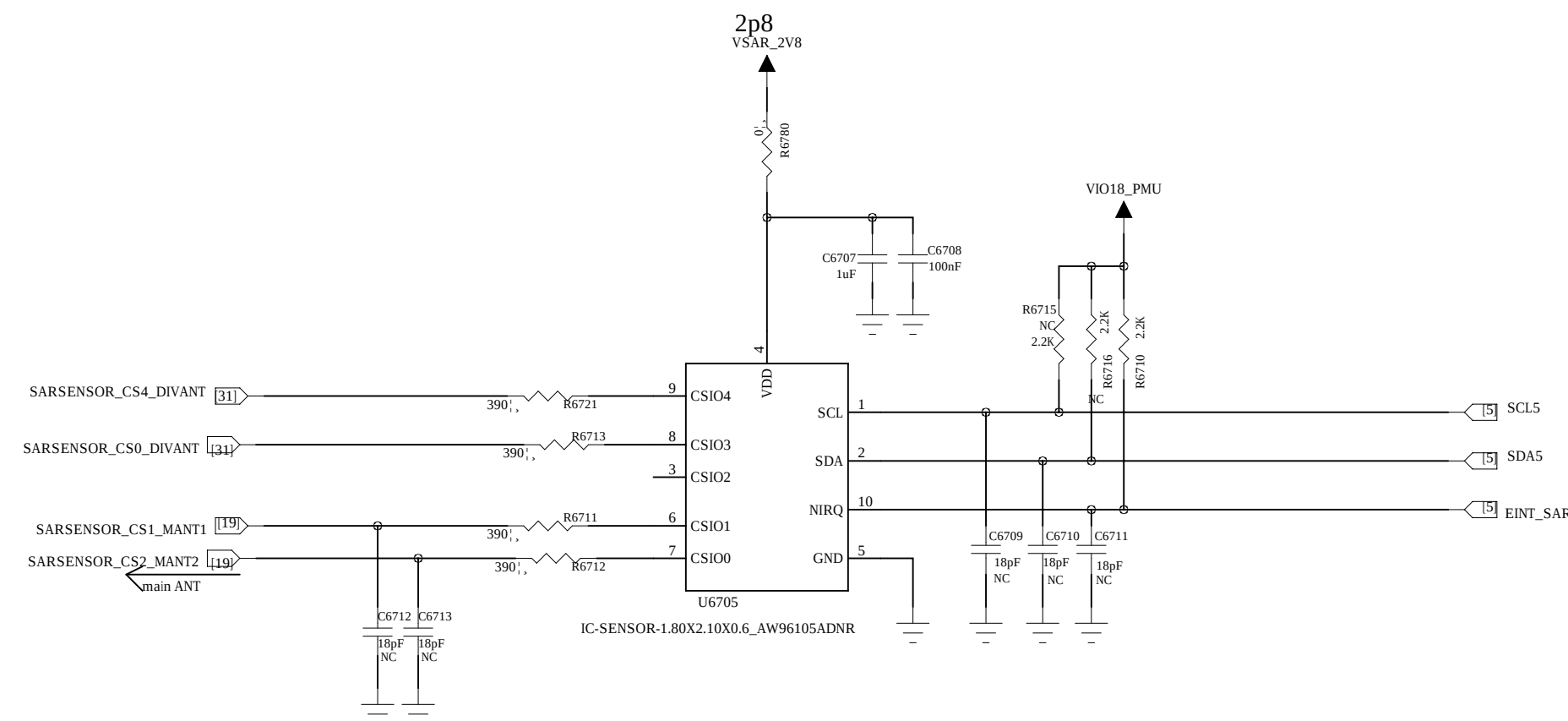
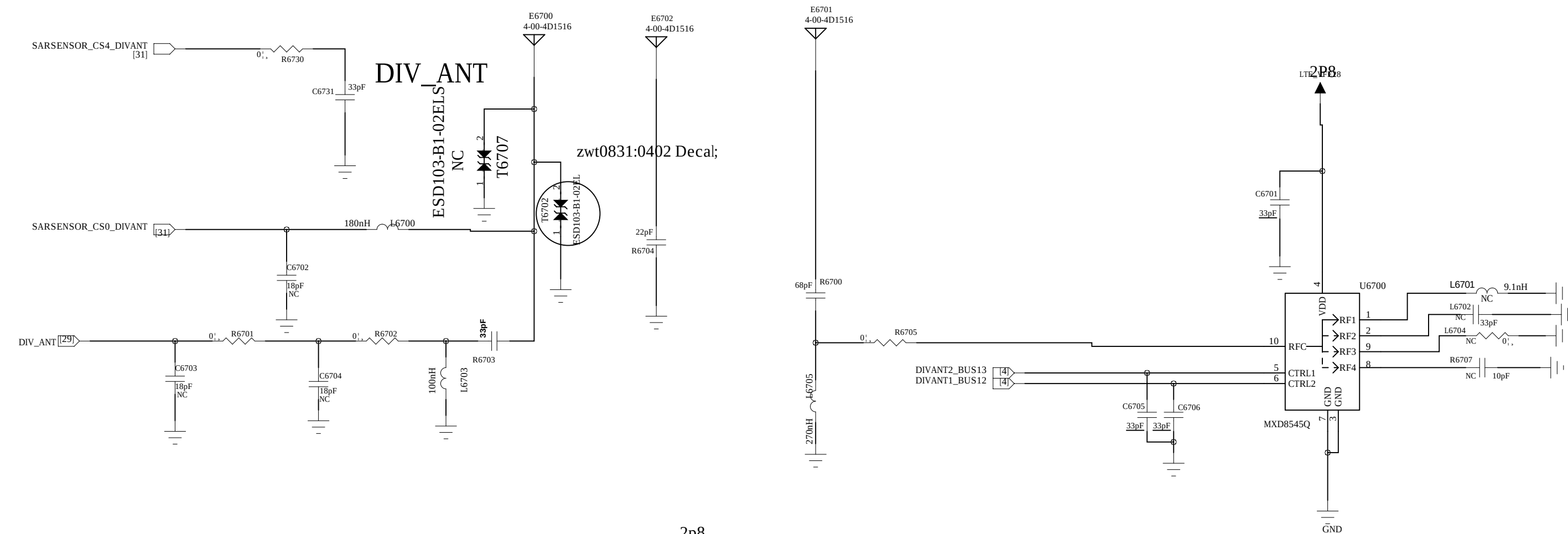
TITLE: MT6765

DRAWN: Marshall	DATED: 2018-09-22A
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

CODE:	SIZE:	DRAWING NO:	REV:
P410	D	P410	V1.0

SCALE: <Scale> SHEET: 20 33

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



2.8V LDO: RP114K281D

		COMPANY: Tinno					
		TITLE: MT6765					
		DRAWN: Marshall		DATED: 2018-09-22A			
		CHECKED: <Checked By>	DATED: <Checked Date>		CODE: P410	SIZE: D	DRAWING NO: P410
QUALITY CONTROL: <QC By>		DATED: <QC Date>					
RELEASED: <Released By>		DATED: <Release Date>		SCALE: <Scale>			SHEET: 3d 33

